

37MBRAQ

Outdoor Unit Single Zone Ductless System
Sizes 36K, 48K, and 58K

Installation Instructions



Fig. 1 —Sizes 36K



Fig. 2 —Sizes 48K - 58K

NOTES: Read the entire instruction manual before starting the installation.

Images are for illustration purposes only. Actual models may differ slightly.

TABLE OF CONTENTS

	Page
SAFETY CONSIDERATIONS	2
INTRODUCTION	8
ACCESSORIES	8
DIMENSIONS	9
INSTALLATION INSTRUCTIONS	10
CLEARANCES	14
REFRIGERANT CONNECTION INSTRUCTIONS	15
WIRING INSTRUCTIONS	17
CONNECTION DIAGRAMS	18
OUTDOOR UNIT WIRING	19
ELECTRICAL CONNECTIONS	20
WIRING INSTRUCTIONS	20
ELECTRICAL DATA	21
WIRING DIAGRAMS	22
AIR EVACUATION	23
TEST RUN	25
CARE AND MAINTENANCE	25
TROUBLESHOOTING	26
DUCTLESS START-UP CHECKLIST	27

SAFETY CONSIDERATIONS

Installing, starting up, and servicing air-conditioning equipment can be hazardous due to system pressures, electrical components, and equipment location (roofs, elevated structures, etc.).

Only trained, qualified installers and service mechanics should install, start-up, and service this equipment.

Untrained personnel can perform basic maintenance functions such as coil cleaning. All other operations should be performed by trained service personnel.

When working on the equipment, observe precautions in the literature and on tags, stickers, and labels attached to the equipment.

Follow all safety codes. Wear safety glasses and work gloves. Keep a quenching cloth and fire extinguisher nearby when brazing. Use care in handling, rigging, and setting bulky equipment.

Read these instructions thoroughly and follow all warnings or cautions included in literature and attached to the unit. Consult local building codes and National Electrical Code (NEC) for special requirements. Recognize safety information.

This is the safety-alert symbol .

When you see this symbol on the unit and in instructions or manuals, be alert to the potential for personal injury. Understand these signal words: **DANGER**, **WARNING**, and **CAUTION**. These words are used with the safety-alert symbol.

DANGER identifies the most serious hazards which will result in severe personal injury or death.

WARNING signifies hazards which could result in personal injury or death.

CAUTION is used to identify unsafe practices which may result in minor personal injury or product and property damage.

NOTE is used to highlight suggestions which will result in enhanced installation, reliability, or operation.



WARNING

ELECTRICAL SHOCK HAZARD

Failure to follow this warning could result in personal injury or death.

Before installing, modifying, or servicing system, the main electrical disconnect switch must be in the OFF position. There may be more than 1 disconnect switch. Lock out and tag switch with a suitable warning label.



WARNING



EXPLOSION HAZARD

Failure to follow this warning could result in personal injury or death. Before installing, modifying, or servicing system, the main electrical disconnect switch must be in the OFF position. There may be more than 1 disconnect switch. Lock out and tag switch with a suitable warning label.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Do not bury more than 36 in. (914 mm) of refrigerant pipe in the ground. If any section of pipe is buried, there must be a 6 in. (152 mm) vertical rise to the valve connections on the outdoor units. If more than the recommended length is buried, refrigerant may migrate to the cooler buried section during extended periods of system shutdown. This causes refrigerant slugging and could possibly damage the compressor at start-up.



WARNING

This appliance is not intended for use by persons (including children) with reduced physical, sensory or mental capabilities, or lack of experience and knowledge, unless they have been given supervision or instruction concerning use of the appliance by a person responsible for their safety. Children should be supervised to ensure that they do not play with the appliance.



WARNING

Only use the specified wire. If the wire is damaged, it must be replaced by the manufacturer, its service agent or similarly qualified persons in order to avoid a hazard. The product must be properly grounded at the time of installation, or electric shock may occur.

For all electrical work, follow all local and national wiring standards, regulations, and the Installation Manual. Connect the cables tightly, and clamp them securely to prevent external forces from damaging the terminal. Improper electrical connections can overheat and cause fire, and may also cause shock. All electrical connections must be made according to the Electrical Connection Diagram located on the panels of the indoor and outdoor units.

All wiring must be properly arranged to ensure that the control board cover can close properly. If the control board cover is not closed properly, it can lead to corrosion and cause the connection points on the terminal to heat up, catch fire, or cause electrical shock.

Disconnection must be incorporated in the fixed wiring in accordance with NEC, CSA, and Local Codes. Do not share the electrical outlet with other appliances. Improper or insufficient power supply can cause fire or electric shock.

If connecting power to fixed wiring, an all-pole disconnection device which has at least 3mm clearances in all poles, and have a leakage current that may exceed 10mA, the residual current device (RCD) having a rated residual operating current not exceeding 30mA, and disconnection must be incorporated in the fixed wiring in accordance with NEC, CSA, and Local Codes.

**WARNING**

Turn off the unit and disconnect the power before performing any installation or repairing. Failure to do so can cause electric shock.

Installation must be performed by an authorized dealer or specialist. Defective installation can cause water leakage, electrical shock, or fire.

Installation must be performed according to the installation instructions. Improper installation can cause water leakage, electrical shock, or fire.

Contact an authorized service technician for repair or maintenance of this unit. This appliance shall be installed in accordance with national wiring regulations. Only use the included accessories, parts, and specified parts for installation. Using non-standard parts can cause water leakage, electrical shock, fire, and can cause the unit to fail.

Install the unit in a firm location that can support the unit's weight. If the chosen location cannot support the unit's weight, or the installation is not done properly, the unit may drop and cause serious injury and damage.

Install drainage piping according to the instructions in this manual. Improper drainage may cause water damage to your home and property.

For units that have an auxiliary electric heater, do not install the unit within 3 feet (1 meter) of any combustible materials.

If combustible gas accumulates around the unit, it may cause fire.

Do not turn on the power until all work has been completed.

When moving or relocating the unit, consult experienced service technicians for disconnection and re-installation of the unit.

How to install the appliance to its support, please read the information for details in "indoor unit installation" and "outdoor unit installation" sections.

NOTE: The air conditioner's circuit board (PCB) is designed with a fuse to provide overcurrent protection. The specifications of the fuse are printed on the circuit board, for example: T3.15AL/250VAC, T5AL/250VAC, T3.15A/250VAC, T5A/250VAC, T20A/250VAC, T30A/250VAC, etc.

NOTE: Only the blast-proof ceramic fuse can be used.

**WARNING****FOR FLAMMABLE REFRIGERANTS**

Do not use means to accelerate the defrosting process or to clean, other than those recommended by the manufacturer.

The appliance shall be stored in a room without continuously operating ignition sources (for example: open flames, an operating gas appliance or an operating electric heater).

Do not pierce or burn. Be aware that refrigerants may not contain an odor.

**WARNING****PERSONAL INJURY AND PROPERTY DAMAGE HAZARD**

For continued performance, reliability, and safety, the only approved accessories and replacement parts are those specified by the equipment manufacturer. The use of non-manufacturer approved parts and accessories could invalidate the equipment limited warranty and result in fire risk, equipment malfunction, and failure. Review the manufacturer's instructions and replacement parts catalogs available from your equipment supplier.

WARNING - RISK OF FIRE DUE TO FLAMMABLE REFRIGERANT USED. FOLLOW HANDLING INSTRUCTIONS CAREFULLY IN COMPLIANCE WITH NATIONAL REGULATIONS.

R-454B



Refrigerant
Safety Group
A2L

R-454B

NOTE: Risk of Fire. Flammable refrigerant used. To be repaired only by trained service personnel. Do not puncture refrigerant tubing.

Table 1 — A (min)

Ho, release height ft (m)

MC or Mrel lbs (kg)	≤ 7.2 (2.2)	7.5 (2.3)	7.9 (2.4)	8.5 (2.6)	9.2 (2.8)	9.8 (3.0)
	≤ 3.91 (1.776)	12 (1.10)				
4.0 (1.8)	60 (5.53)	57 (5.29)	55 (5.07)	50 (4.68)	47 (4.34)	44 (4.05)
4.4 (2.0)	66 (6.14)	63 (5.88)	61 (5.63)	56 (5.2)	52 (4.83)	48 (4.5)
4.9 (2.2)	73 (6.76)	70 (6.46)	67 (6.19)	62 (5.72)	57 (5.31)	53 (4.95)
5.3 (2.4)	79 (7.37)	76 (7.05)	73 (6.76)	67 (6.24)	62 (5.79)	58 (5.41)
5.7 (2.6)	86 (7.99)	82 (7.64)	79 (7.32)	73 (6.76)	68 (6.27)	63 (5.86)
6.2 (2.8)	93 (8.6)	89 (8.23)	85 (7.88)	78 (7.28)	73 (6.76)	68 (6.31)
6.6 (3.0)	99 (9.21)	95 (8.81)	91 (8.45)	84 (7.8)	78 (7.24)	73 (6.76)
7.1 (3.2)	106 (9.83)	101 (9.4)	97 (9.01)	90 (8.32)	83 (7.72)	78 (7.21)
7.5 (3.4)	112 (10.44)	108 (9.99)	103 (9.57)	95 (8.84)	88 (8.2)	82 (7.66)
7.9 (3.6)	119 (11.06)	114 (10.58)	109 (10.14)	101 (9.36)	94 (8.69)	87 (8.11)
8.4 (3.8)	126 (11.67)	120 (11.16)	115 (10.7)	106 (9.88)	99 (9.17)	92 (8.56)
8.8 (4.0)	132 (12.29)	126 (11.75)	121 (11.26)	112 (10.4)	104 (9.65)	97 (9.01)
9.3 (4.2)	139 (12.9)	133 (12.34)	127 (11.82)	117 (10.91)	109 (10.14)	102 (9.46)
9.7 (4.4)	145 (13.51)	139 (12.93)	133 (12.39)	123 (11.43)	114 (10.62)	107 (9.91)
10.1 (4.6)	152 (14.13)	145 (13.51)	139 (12.95)	129 (11.95)	119 (11.1)	112 (10.36)
10.6 (4.8)	159 (14.74)	152 (14.1)	145 (13.51)	134 (12.47)	125 (11.58)	116 (10.81)
11 (5.0)	165 (15.36)	158 (14.69)	152 (14.08)	140 (12.99)	130 (12.07)	121 (11.26)
11.5 (5.2)	172 (15.97)	164 (15.28)	158 (14.64)	145 (13.51)	135 (12.55)	126 (11.71)
11.9 (5.4)	179 (16.58)	171 (15.86)	164 (15.2)	151 (14.03)	140 (13.03)	131 (12.16)
12.3 (5.6)	185 (17.2)	177 (16.45)	170 (15.77)	157 (14.55)	145 (13.51)	136 (12.61)
12.8 (5.8)	192 (17.81)	183 (17.04)	176 (16.33)	162 (15.07)	151 (14)	141 (13.06)
13.2 (6.0)	198 (18.43)	190 (17.63)	182 (16.89)	168 (15.59)	156 (14.48)	145 (13.51)
13.7 (6.2)	205 (19.04)	196 (18.21)	188 (17.45)	173 (16.11)	161 (14.96)	150 (13.96)
14.1 (6.4)	212 (19.66)	202 (18.8)	194 (18.02)	179 (16.63)	166 (15.44)	155 (14.41)
14.6 (6.6)	218 (20.27)	209 (19.39)	200 (18.58)	185 (17.15)	171 (15.93)	160 (14.86)
15 (6.8)	225 (20.88)	215 (19.98)	206 (19.14)	190 (17.67)	177 (16.41)	165 (15.32)
15.4 (7.0)	231 (21.5)	221 (20.56)	212 (19.71)	196 (18.19)	182 (16.89)	170 (15.77)
15.9 (7.2)	238 (22.11)	228 (21.15)	218 (20.27)	201 (18.71)	187 (17.37)	175 (16.22)

Amin is the required minimum room area in ft²/m².**mc** is the actual refrigerant charge in the system in lbs (kg).**mREL** is the refrigerant releasable charge in lbs (kg).**hinst** is the height of the bottom of the unit relative to the floor of the room after installation.**Ho** is the vertical distance in feet (meters) from the floor to the point of release when the appliance is installed.**Ho**, release height = install height**WARNING**

The minimum room area or minimum room area of conditioned space is based on releasable charge or total system refrigerant charge.

For R-454B Refrigerant Charge Amount and Minimum Room Area:

The indoor and outdoor units are designed to be used together. Please check the unit you purchased: The height of the room cannot be less than 7.3 feet / 2.2 meters; and the minimum room area of operating or storage should be specified in Table 1.

**WARNING****PRIOR TO INSTALLATION**

Before installing the indoor unit, ensure the compatibility with the outdoor unit using the product data as a reference. It is also necessary to confirm the proper application of the equipment and to perform a heat load calculation for proper sizing.

1. Installation (where refrigerant pipes are allowed)

Any person who is involved with working on or breaking into a refrigerant circuit should hold a current valid certificate from an industry-accredited assessment authority, which authorizes their competence to handle refrigerants safely in accordance with an industry recognized assessment specification.

Maintenance and repair requiring the assistance of other skilled personnel shall be carried out under the supervision of the person competent in the use of flammable refrigerants.

That the installation of pipe-work shall be kept to a minimum.

That pipe-work shall be protected from physical damage.

Where refrigerant pipes shall be compliance with national gas regulations.

That mechanical connections shall be accessible for maintenance purposes.

Be more careful that foreign matter (oil, water, etc) does not enter the piping. Also, when storing the piping, securely seal the opening by pinching, taping, etc.

All working procedure that effects safety means shall only be carried by competent persons.

Appliance shall be stored in a well ventilated area where the room size corresponds to the room area as specific for operation.

Joints shall be tested with detection equipment with a capability of 1/8 oz (5grams)/year of refrigerant or better, with the equipment in standstill and under operation or under a pressure of at least these standstill or operation conditions after installation.

In cases that require mechanical ventilation, ventilation openings shall be kept clear of obstruction.

LEAK DETECTION SYSTEM installed. Unit must be powered except for service. For the unit with refrigerant sensor, when the refrigerant sensor detects refrigerant leakage, the indoor unit will display a error code and emit a buzzing sound, the compressor of outdoor unit will immediately stop, and the indoor fan will start running. The service life of the refrigerant sensor is 15 years. When the refrigerant sensor malfunctions, the indoor unit will display the error code FHCC. The refrigerant sensor can not be repaired and can only be replaced by the manufacture. It shall only be replaced with the sensor specified by the manufacture.

2. When a FLAMMABLE REFRIGERANT is used

The requirements for installation space of appliance and/or ventilation requirements are determined according to:

- the mass charge amount (M) used in the appliance,
- the installation location,
- the type of ventilation of the location or of the appliance.
- piping material, pipe routing, and installation shall include protection from physical damage in operation and service, and be in compliance with national and local codes and standards, such as ASHRAE 15 or CSA B52. All field joints shall be accessible for inspection prior to being covered or enclosed.
- that protection devices, piping, and fittings shall be protected as far as possible against adverse environmental effects, for example, the danger of water collecting and freezing in relief pipes or the accumulation of dirt and debris;
- that piping in refrigeration systems shall be so designed and installed to minimize the likelihood of hydraulic shock damaging the system;
- that steel pipes and components shall be protected against corrosion with a rustproof coating before applying any insulation;
- that precautions shall be taken to avoid excessive vibration or pulsation;
- the minimum floor area of the room shall be mentioned in the form of a table or a single figure without reference to a formula;

–after completion of field piping for split systems, the field pipework shall be pressure tested with an inert gas and then vacuum tested prior to refrigerant charging, according to the following requirements:

- a. The required nitrogen pressure is 500 psi
 - b. The test pressure after removal of pressure source shall be maintained for at least 1 hour with no decrease of pressure indicated by the test gauge, with test gauge resolution not exceeding 5% of the test pressure.
 - c. During the evacuation test, after achieving a vacuum level specified in the manual or less, the refrigeration system shall be isolated from the vacuum pump and the pressure shall not rise above 1500 microns within 10 min. The vacuum pressure level shall be specified in the manual, and shall be the lessor of 500 microns or the value required for compliance with national and local codes and standards, which may vary between residential, commercial, and industrial buildings.
- Field-made refrigerant joints indoors shall be tightness tested according to the following requirements: The test method shall have a sensitivity of 1/8 oz (5grams) per year of refrigerant or better under a pressure of at least 125% of the maximum allowable pressure. No leak shall be detected.

3. Qualification of Workers

Any maintenance, service and repair operations must be performed by skilled and authorized personnel. Every working procedure that effects safety means shall only be carried out by competent persons that joined the training and achieved competence should be documented by a certificate. The training of these procedures is carried out by national training organizations or manufacturers that are accredited to teach the relevant national competency standards that may be set in legislation. Examples for such working procedures are:

- breaking into the refrigerating circuit;
- opening of sealed components;
- opening of ventilated enclosures.

4. Checks to the area

Prior to beginning work on systems containing flammable refrigerants, safety checks are necessary to ensure that the risk of ignition is minimized. For repair to the refrigerating system, the following precautions shall be complied with prior to conducting work on the system.

5. Work procedure

Works shall be undertaken under a controlled procedure so as to minimize the risk of a flammable gas or vapor being present while the work is being performed.

6. General work area

All maintenance staff and others working in the local area shall be instructed on the nature of work being carried out. work in confined spaces shall be avoided.

7. Checking for presence of refrigerant

The area shall be checked with an appropriate refrigerant detector prior to and during work, to ensure the technician is aware of potentially flammable atmospheres. Ensure that the leak detection equipment being used is suitable for use with flammable refrigerants, i.e. no sparking, adequately sealed or intrinsically safe.

8. Presence of fire extinguisher

If any hot work is to be conducted on the refrigeration equipment or any associated parts, appropriate fire extinguishing equipment shall be available to hand. Have a dry power or CO₂ fire extinguisher adjacent to the charging area.

9. No ignition sources

No person carrying out work in relation to a REFRIGERATING SYSTEM which involves exposing any pipe work shall use any sources of ignition in such a manner that it may lead to the risk of fire or explosion. All possible ignition sources, including cigarette smoking, should be kept sufficiently far away from the site of installation, repairing, removing and disposal, during which refrigerant can possibly be released to the surrounding space. Prior to work taking place, the area around the equipment is to be surveyed to make sure that there are no flammable hazards or ignition risks. "No Smoking" signs shall be displayed.

10. Ventilated area

Ensure that the area is in the open or that it is adequately ventilated before breaking into the system or conducting any hot work. A degree of ventilation shall continue during the period that the work is carried out. The ventilation should safely disperse any released refrigerant and preferably expel it externally into the atmosphere.

11. Checks to the refrigeration equipment

Where electrical components are being changed, they shall be fit for the purpose and to the correct specification. At all times the manufacturer's maintenance and service guidelines shall be followed. If in doubt consult the manufacturer's technical department for assistance. The following checks shall be applied to installations using FLAMMABLE REFRIGERANTS:

- the actual refrigerant charge is in accordance with the room size within which the refrigerant containing parts are installed;
- the ventilation machinery and outlets are operating adequately and are not obstructed;
- if an indirect refrigerating circuit is being used, the secondary circuits shall be checked for the presence of refrigerant;
- marking to the equipment continues to be visible and legible, marking and signs that are illegible shall be corrected;
- refrigeration pipe or components are installed in a position where they are unlikely to be exposed to any substance which may corrode refrigerant containing components, unless the components are constructed of materials which are inherently resistant to being corroded or are suitably protected against being so corroded.

12. Checks to electrical devices

Repair and maintenance to electrical components shall include initial safety checks and component inspection procedures. If a fault exists that could compromise safety, then no electrical supply shall be connected to the circuit until it is satisfactorily dealt with. If the fault cannot be corrected immediately but it is necessary to continue operation, and adequate temporary solution shall be used. This shall be reported to the owner of the equipment so all parties are advised. Initial safety checks shall include:

- that capacitors are discharged: this shall be done in a safe manner to avoid possibility of sparking
- that there no live electrical components and wiring are exposed while charging, recovering or purging the system;
- that there is continuity of earth bonding.

13. Cabling

Check that cabling is not subjected to wear, corrosion, excessive pressure, vibration, sharp edges or any other adverse environmental effects. The check shall also take into account the effects of aging or continual vibration from sources such as compressors or fans.

14. Detection of flammable refrigerants

Under no circumstances shall potential sources of ignition be used in the searching for or detection of refrigerant leaks. A halide torch (or any other detector using a naked flame) shall not be used.

The following leak detection methods are deemed acceptable for refrigerant systems. Electronic leak detectors that have a sensitivity of 1/8 oz (5g)/year may be used to detect leaks of flammable refrigerants. (Detection equipment shall be calibrated in a refrigerant-free area.) Ensure that the detector is not a potential source of ignition and is suitable for the refrigerant used. Leak detection equipment shall be set at a percentage of the LFL of the refrigerant and shall be calibrated to the refrigerant employed, and the appropriate percentage of gas (25% maximum) is confirmed. Leak detection fluids are also suitable for use in external leak detection.

NOTE: Examples of leak detection fluids are:

- bubble method,
- fluorescent method agents.

If a leak is suspected, all open flames shall be removed/extinguished.

If a leakage of refrigerant is found which requires brazing, all of the refrigerant shall be recovered from the system, or isolated (by means of shut off valves) in a part of the system remote from the leak. See the following instructions of removal of refrigerant.

15. Removal and Evacuation

When breaking into the refrigerant circuit to make repairs or for any other purpose, conventional procedures shall be used. However, for flammable refrigerants it is important that best practice be followed, since flammability is a consideration. The following procedure shall be adhered to:

- safely remove refrigerant following local and national regulations; evacuate
- purge the circuit with nitrogen
- evacuate (requirement)
- continuously flush or purge with nitrogen when using flame to open circuit
- open the circuit

The refrigerant charge shall be recovered into the correct recovery cylinders. Charging must be performed by liquid charging method. For appliances containing flammable refrigerants, the system shall be purged with oxygen-free nitrogen to render the appliance safe for flammable refrigerants. This process might need to be repeated several times. Compressed air or oxygen shall not be used for purging refrigerant systems.

For appliances containing flammable refrigerants, refrigerants purging shall be achieved by breaking the vacuum in the system with oxygen-free nitrogen and continuing to fill until the working pressure is achieved, then venting to atmosphere, and finally pulling down to a vacuum (requirement). This process shall be repeated until no refrigerant is within the system (requirement). When the final oxygen-free nitrogen charge is used, the system shall be vented down to atmospheric pressure to enable work to take place. The outlet for the vacuum pump shall not be close to any potential ignition sources, and ventilation shall be available.

16. Charging procedures

In addition to conventional charging procedures, the following requirements shall be followed:

- Work shall be undertaken with appropriate tools only (In case of uncertainty, consult the manufacturer of the tools for use with flammable refrigerants). Ensure that contamination of different refrigerants does not occur when using charging equipment.

Hoses or lines shall be as short as possible to minimize the amount of refrigerant contained in them.

- Cylinders shall be kept upright.
- Ensure that the refrigeration system is earthed prior to charging the system with refrigerant.
- Label the system when charging is complete (if not already).
- Extreme care shall be taken not to overfill the refrigeration system.
- Prior to recharging the system it shall be pressure tested with oxygen free nitrogen (OFN). The system shall be leak tested on completion of charging but prior to commissioning. A follow up leak test shall be carried out prior to leaving the site.

17. Decommissioning

Before carrying out this procedure, it is essential that the technician is completely familiar with the equipment and all its detail. It is recommended good practice that all refrigerants are recovered safely. Prior to the task being carried out, an oil and refrigerant sample shall be taken in case analysis is required prior to re-use of recovered refrigerant. It is essential that electrical power is available before the task is commenced.

- a. Become familiar with the equipment and its operation.
- b. Isolate system electrically
- c. Before attempting the procedure ensure that:
 - mechanical handling equipment is available, if required, for handling refrigerant cylinders;
 - all personal protective equipment is available and being used correctly;
 - the recovery process is supervised at all times by a competent person;
 - recovery equipment and cylinders conform to the appropriate standards.
- d. Pump down refrigerant system, if possible.
- e. If a vacuum is not possible, make a manifold so that refrigerant can be removed from various parts of the system.
- f. Make sure that cylinder is situated on the scales before recovery takes place.
- g. Start the recovery machine and operate in accordance with instructions.
- h. Do not overfill cylinders (no more than 80 % volume liquid charge)
- i. Do not exceed the maximum working pressure of the cylinder, even temporarily.
- j. When the cylinders have been filled correctly and the process completed, make sure that the cylinders and the equipment are removed from site promptly and all isolation valves on the equipment are closed.
- k. Recovered refrigerant shall not be charged into another refrigeration system unless it has been cleaned and checked.

18. Labeling

Equipment shall be labeled stating that it has been de-commissioned and emptied of refrigerant. The label shall be dated and signed. For appliances containing FLAMMABLE REFRIGERANTS, ensure that there are labels on the equipment stating the equipment contains FLAMMABLE REFRIGERANT.

19. Recovery

When removing refrigerant from a system, either for servicing or decommissioning, it is recommended good practice that all refrigerants are removed safely.

When transferring refrigerant into cylinders, ensure that only appropriate refrigerant recovery cylinders are employed. Ensure that the correct number of cylinders for holding the total system charge is available. All cylinders to be used are designated for the recovered refrigerant and labelled for that refrigerant (i. e. special cylinders for

the recovery of refrigerant). Cylinders shall be complete with pressure-relief valve and associated shut-off valves in good working order. Empty recovery cylinders are evacuated and, if possible, cooled before recovery occurs.

The recovery equipment shall be in good working order with a set of instructions concerning the equipment that is at hand and shall be suitable for the recovery of the flammable refrigerant. If in doubt? the manufacturer should be consulted. In addition, a set of calibrated weighing scales shall be available and in good working order. Hoses shall be complete with leak-free disconnect couplings and in good condition.

The recovered refrigerant shall be processed according to local legislation in the correct recovery cylinder, and the relevant waste transfer note arranged. Do not mix refrigerants in recovery units and especially not in cylinders.

If compressors or compressor oils are to be removed, ensure that they have been evacuated to an acceptable level to make certain that flammable refrigerant does not remain within the lubricant. The compressor body shall not be heated by an open flame or other ignition sources to accelerate this process. When oil is drained from a system, it shall be carried out safely.

20. Unventilated areas

For appliances containing more than for any refrigerating circuit, the manual shall include a statement advising that an unventilated area where the appliance using FLAMMABLE REFRIGERANTS is installed shall be so constructed that should any refrigerant leak, it will not stagnate so as to create a fire or explosion hazard. This shall include:

- a warning that if appliances with A2L REFRIGERANTS connected via an air duct system to one or more rooms are installed in a room with an area less than $>A_{min}$ as determined in Clause GG.2, that room shall be without continuously operating open flames (for example an operating gas appliance) or other POTENTIAL IGNITION SOURCES (for example an operating electric heater, hot surfaces). A flame-producing device may be installed in the same space if the device is provided with an effective flame arrest;
- for appliances using A2L REFRIGERANTS connected via an air duct system to one or more rooms, a warning with the substance of the following: "Auxiliary devices which may be a POTENTIAL IGNITION SOURCE shall not be installed in the duct work. Examples of such POTENTIAL IGNITION SOURCES are hot surfaces with a temperature exceeding X °C and electric switching devices".

NOTE: X is the maximum allowable surface temperature as defined in 22.117.

The manufacturer should specify other potential continuously operating sources known to cause ignition of the refrigerant used.

The appliance shall be stored so as to prevent mechanical damage from occurring.

- for appliances using A2L refrigerants connected via an air duct system to one or more rooms, a warning that only auxiliary devices approved by the appliance manufacturer or declared suitable with the refrigerant shall be installed in connecting ductwork. The manufacturer shall list in the instructions all approved auxiliary devices by manufacturer and model number for use with the specific appliance, if those devices have a potential to become an ignition source.
- a warning that if appliances connected via an air duct system to one or more rooms with A2L REFRIGERANTS are installed in a room with an area less than 4m² as determined in Clause GG.2. or installed in a room with an EFFECTIVE DISPERSAL VOLUME VED less than the minimum as determined by Clause 101.DVN.8, that room shall be without continuously operating open flames (e.g.

an operating gas appliance) or other POTENTIAL IGNITION SOURCES (for e.g. an operating electric heater, hot surfaces). A flame-producing device may be installed in the same space if the device is provided with an effective flame arrest.

- for REFRIGERANT DETECTION SYSTEMS, the function and operation and required servicing measures;
- for LIMITED LIFE REFRIGERANT SENSORS Used in REFRIGERANT DETECTION SYSTEMS, the specified end-of-life and replacement instructions;
- REFRIGERANT SENSORS for REFRIGERANT DETECTION SYSTEMS Shall Only be replaced with sensors specified by the appliance manufacture; and instructions to verify actuation of mitigation actions per Annex GG or Annex 101.DVN as applicable.

For appliances using FLAMMABLE REFRIGERANTS with safety features that depend upon the proper function of a leak detection system used for leak mitigation, the instructions and unit markings shall contain the substance of the following:

"LEAK DETECTION SYSTEM installed. Unit must be powered except for service."

If any remote located REFRIGERANT SENSOR is employed to detect leaked refrigerant, such a remote located REFRIGERANT SENSOR shall also apply to this marking or be accompanied by such instructions.

21. Transportation, marking and storage for units that employ flammable refrigerants
- a. General

The following information is provided for units that employ FLAMMABLE REFRIGERANTS.

- b. Transport of equipment containing flammable refrigerants

Attention is drawn to the fact that additional transportation regulations may exist with respect to equipment containing flammable gas. The maximum number of pieces of equipment or the configuration of the equipment permitted to be transported together will be determined by the applicable transport regulations.

- c. Marking of equipment using signs

Signs for similar appliances used in a work area are generally addressed by local regulations and give the minimum requirements for the provision of safety and/or health signs for a work location.

All required signs are to be maintained and employers should ensure that employees receive suitable and sufficient instruction and training on the meaning of appropriate safety signs and the actions that need to be taken in connection with these signs.

The effectiveness of signs should not be diminished by too many signs being placed together.

Any pictograms used should be as simple as possible and contain only essential details.

- d. Disposal of equipment using flammable refrigerants

See national regulations.

- e. Storage of equipment/appliances

The storage of the appliance should be in accordance with the applicable regulations or instructions, whichever is more stringent.

- f. Storage of packed (unsold) equipment

Storage package protection should be constructed in such a way that mechanical damage to the equipment inside the package will not cause a leak of the REFRIGERANT CHARGE.

The maximum number of pieces of equipment permitted to be stored together will be determined by local regulations.

	WARNING	This symbol shows that this appliance used a flammable refrigerant. If the refrigerant is leaked and exposed to an external ignition source, there is a risk of fire.
	CAUTION	This symbol shows that the operation manual should be read carefully.
	CAUTION	This symbol shows that a service personnel should be handling this equipment with reference to the installation manual.
	CAUTION	This symbol shows that information is available such as the operating manual or installation manual.
	CAUTION	This symbol shows that information is available such as the operating manual or installation manual.

INTRODUCTION

The Horizontal Discharge Outdoor units are R-454b condensing units designed with application flexibility in mind.

These units have a max total piping length up to 246ft (75m) and a maximum piping lift of up to 98 ft (30m).

The inverter driven compressor is designed to run at various input power frequencies (Hz) which controls the compressor's motor speed.

ACCESSORIES

The system is shipped with the following accessories. Use all of the installation parts and accessories to install the system. Improper installation may result in water leakage, electrical shock and fire, or cause the equipment to fail. Keep the installation manual in a safe place and do not discard any other accessories until the installation work has been completed.

Table 2 — Accessories

PART NO.	PART NAME	QTY.
1	Outdoor unit	1
-	Literature package including installation instructions	1
-	Grommet to secure the outdoor unit (helps with vibration prevention during unit operation)	4
-	Drain Joint	1
-	Drain Hose	1

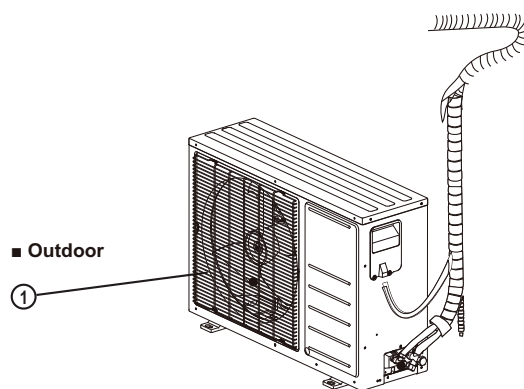


Fig. 3 —Parts List

NOTES:

- If the outdoor unit is higher than the indoor unit, prevent rain from flowing into the indoor unit along the connection pipe by making a downward arc in the connection pipe before it enters the wall to the indoor unit. This ensures that rain drips from the connection pipe before it enters the wall.
- Piping and the interconnecting wiring are field supplied.
- The illustration above is only a sketch. Different models may be slightly different.

Table 3 lists the units covered in this manual.

Table 3 — Unit Sizes

SYSTEM TONS	kBTUh	VOLTAGE - PHASE	OUTDOOR MODEL
3	36,000	208/230 - 1	37MBRAQ36AA3
4	48,000	208/230 - 1	37MBRAQ48AA3
5	58,000	208/230 - 1	37MBRAQ58AA3

DIMENSIONS

Table 4 — Dimensions and Weights

System Size		36K	48K	58K
		(208/230 V)	(208/230 V)	(208/230 V)
Height (H)	in (mm)	31.89 (810)	52.48 (1333)	52.48 (1333)
Width (W)	in (mm)	37.24 (946)	37.48 (952)	37.48(952)
Depth (D)	in (mm)	16.14 (410)	16.34 (415)	16.34(415)
Weight -Net	lbs. (kg)	152.34 (69.1)	208.78(94.7)	209.44 (95)

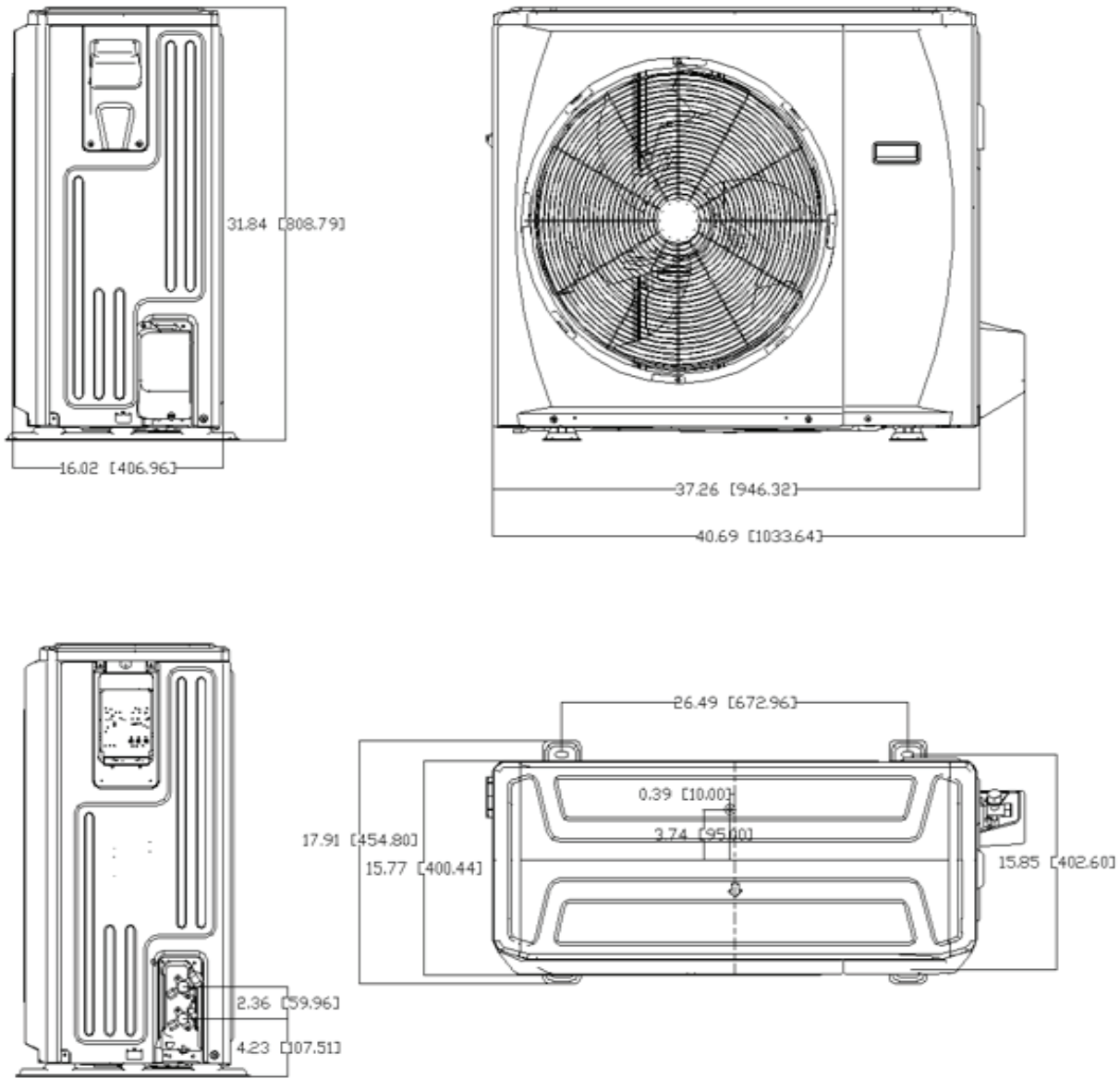


Fig. 4 —Size 36K

DIMENSIONS (CONT.)

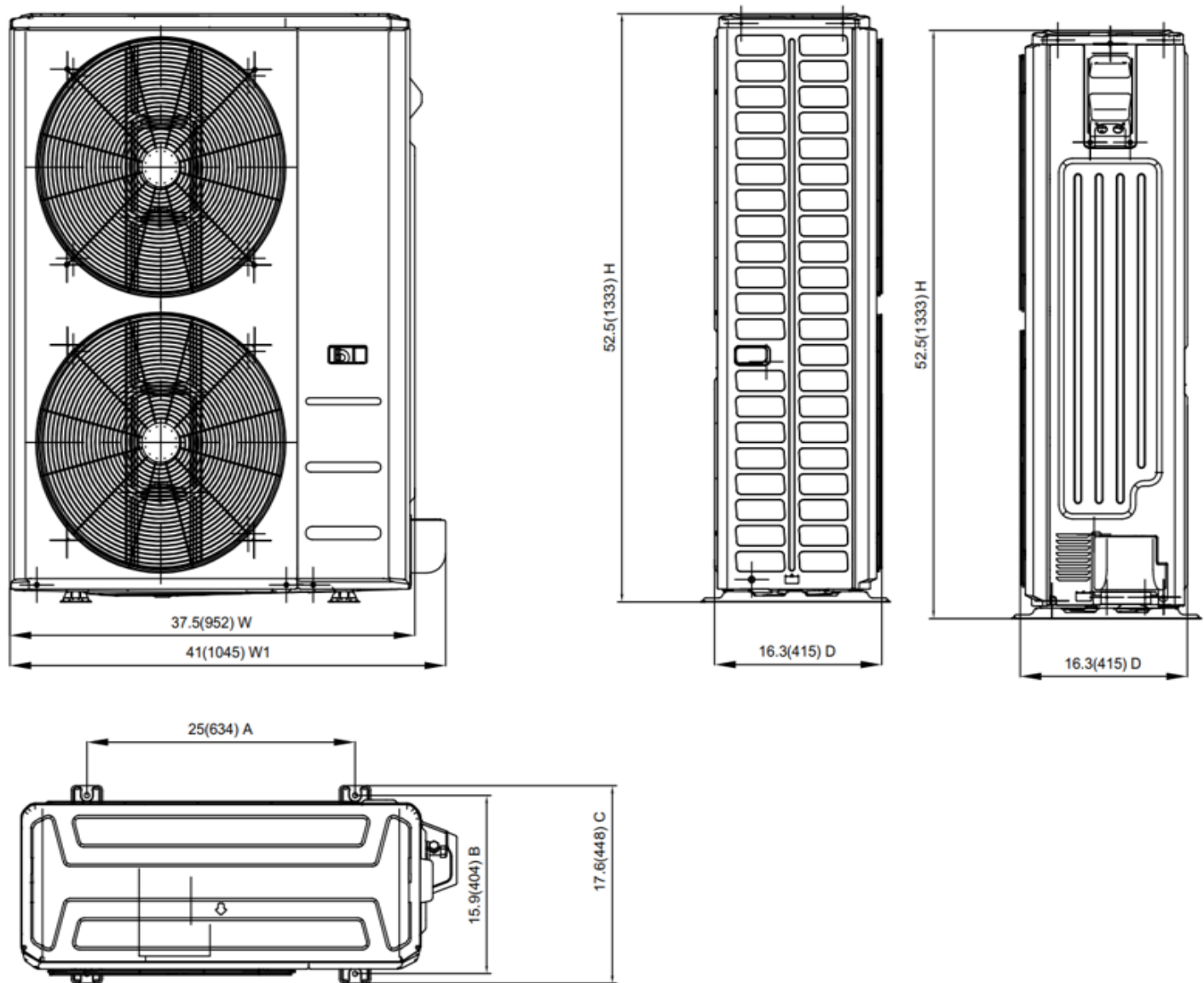


Fig. 5 — Sizes 48K - 58K

INSTALLATION INSTRUCTIONS

- A location which is convenient to installation and not exposed to strong winds.
- A location which can bear the weight of the outdoor unit and where the outdoor unit can be mounted in a level position.
- A location which provides appropriate clearances (see Fig. 6).
- Do not install the indoor or outdoor units in a location with special environmental conditions. For those applications, contact your Sales representative.
- Allow sufficient space for airflow and service of the unit. See Fig. 6 — on page 12 for the required minimum distances between the unit or walls.

NOTE: DO NOT install the indoor or outdoor units in a location with special environmental conditions. For those applications, contact your Ductless representative.



WARNING

PRODUCT INSTALLATION

- Installation must be performed by an authorized dealer or specialist. A defective installation can cause water leakage, electrical shock, or fire.
- The installation must be performed according to the installation instructions. Improper installation can cause water leakage, electrical shock, or fire. (In North America, installation must be performed in accordance with the requirements of NEC or CEC by authorized personnel only.)
- Contact an authorized service technician for repair or maintenance of this unit. This appliance must be installed in accordance with local codes.
- Only use the included accessories, parts, and specified parts for installation. Using non-standard parts can cause water leakage, electrical shock, fire, or unit failure.
- Install drainage piping according to the instructions in this manual. Improper drainage may cause water damage to your home and property.
- **DO NOT** install the unit in a location that may be exposed to combustible gas leaks. If combustible gas accumulates around the unit, it may cause a fire.
- **DO NOT** turn on the power until all work has been completed.
- When moving or relocating the system, consult experienced service technicians for the disconnection and re-installation of the unit.

NOTE: Basepan built-in with multiple holes for proper draining during defrost. For applications where it is required to seal these holes and re-direct the condensation drain, rubber plugs are available through RCD.

Table 5 — Base pan Rubber Plugs

OUTDOOR UNIT MODEL NUMBER PER UNIT	BASE PAN RUBBER PLUGS RCD PART NUMBER	QTY.
36K	12600801A00117	5
48K	12600801A00118	5
58K	12600801A00118	5

IMPORTANT: Both refrigerant lines must be insulated separately. Use refrigerant grade tubing ONLY. No other type tubing may be used. Use of other types of tubing will void the manufacturer's warranty.

- The minimum refrigerant line length between the indoor and outdoor units is 10 ft. (3m).
- When paired with Ductless indoor units, size the line sets based on the connection size of the indoor unit unless specified.
- When paired with conventional or multi-family fan coils, size the line sets based on the liquid and gas connection size of the outdoor unit; a field supplied reducer may be required.
- All outdoor units have an electronic expansion valve to manage the refrigerant flow of the fan coil connected.
- Do not open the service valves or remove the protective caps from the tubing ends until all connections are made.
- Bend the tubing with bending tools to avoid kinks and flat spots.
- Keep the tubing free of dirt, sand, moisture, and other contaminants to avoid damaging the refrigerant system.
- Avoid sags in the suction line to prevent the formation of oil traps.
- Insulate each tube with a minimum 3/8-in. (10 mm) wall thermal pipe insulation. Inserting the tubing into the insulation before making the connections saves time and improves installation quality.

WARNING - RISK OF FIRE DUE TO FLAMMABLE REFRIGERANT USED. FOLLOW HANDLING INSTRUCTIONS CAREFULLY IN COMPLIANCE WITH NATIONAL REGULATIONS.

R-454B



Refrigerant
Safety Group
A2L

R-454B



WARNING

- All field piping must be completed by a licensed technician and must comply with the local and national regulations.
- When the system is installed in a small room, measures must be taken to prevent the refrigerant concentration in the room from exceeding the safety limit in the event of refrigerant leakage. If the refrigerant leaks and its concentration exceeds its proper limit, hazards due to lack of oxygen may result.
- When installing the refrigeration system, ensure that air, dust, moisture or foreign substances do not enter the refrigerant circuit. Contamination in the system may cause poor operating capacity, high pressure in the refrigeration cycle, explosion or injury. Ventilate the area immediately if there is refrigerant leakage during the installation.
- Leaked refrigerant gas is hazardous. Ensure there is no refrigerant leakage after completing the installation work.



CAUTION

DO NOT install the connecting pipe until both the indoor and outdoor units have been installed.

Insulate both the gas and liquid piping to prevent condensation.

INSTALLATION SUMMARY

A. INSTALLATION INSTRUCTIONS

(Set Installation Location, Install Drain Joint, Anchor Outdoor Unit).

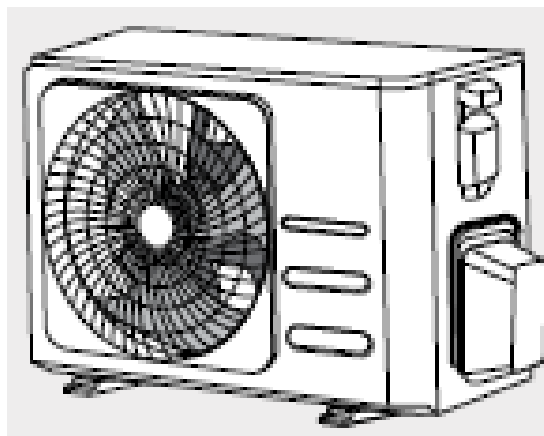


Fig. 6 —Install the Outdoor Unit

B. REFRIGERANT CONNECTION INSTRUCTIONS

(Cut pipes, Remove burrs, etc.).

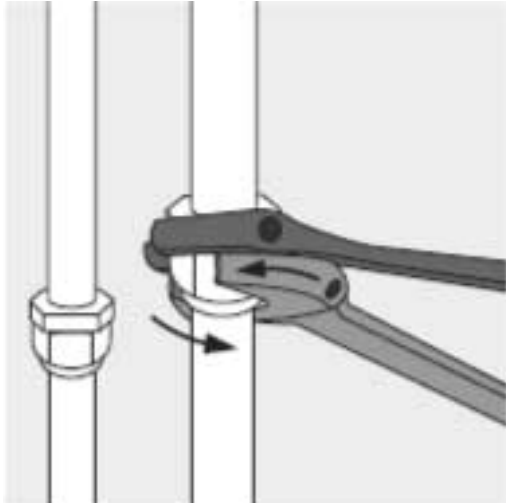


Fig. 7 —Connect the Refrigerant Pipes

C. WIRING INSTRUCTIONS

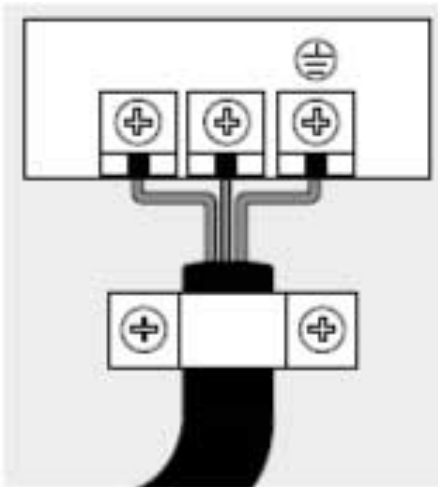


Fig. 8 —Connect the Wires

D. AIR EVACUATION

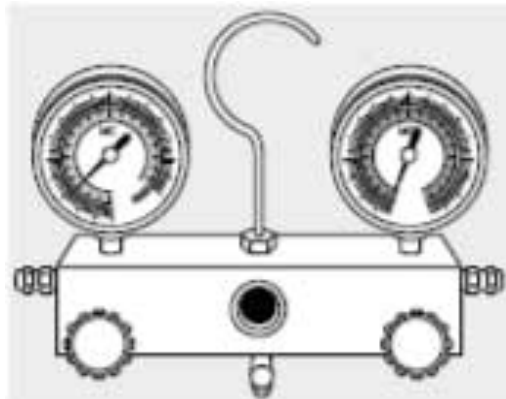


Fig. 9 —Evacuate the Refrigeration System

E. TEST RUN

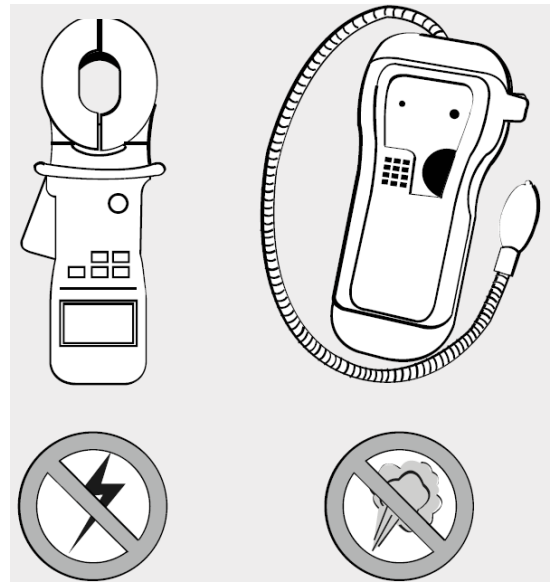


Fig. 10 —Perform a Test Run

22.Install drain joint

Step 1: Find out the base pan hole of outdoor unit.

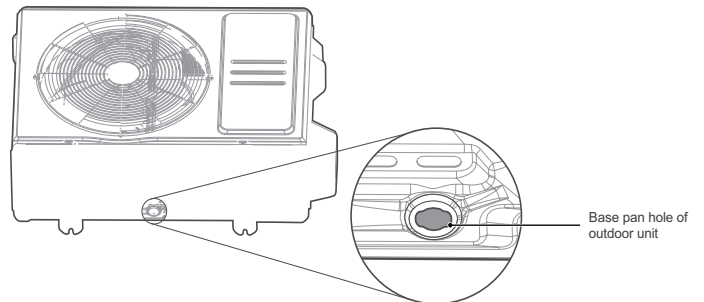


Fig. 11 —Find Base Pan Hole

Step 2:

- Fit the rubber seal on the end of the drain joint that will connect to the outdoor unit.
- Insert the drain joint into the hole in the base pan of the unit. The drain joint will click in place.
- Connect a drain hose extension (not included) to the drain joint to redirect water from the unit during heating mode.

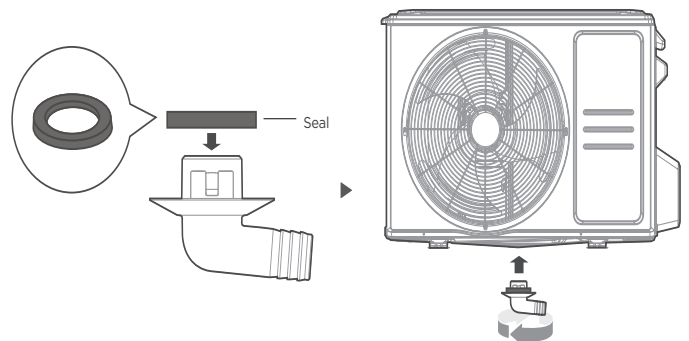


Fig. 12 —Connect Drain Hose to Drain Joint



CAUTION

IN COLD CLIMATES

Ensure the drain hose is as vertical as possible to ensure swift water drainage. If water drains too slowly, it can freeze in the hose and flood the unit.

23.Anchor Outdoor Unit



WARNING

When drilling into concrete, eye protection is recommended at all times.

- The outdoor unit can be anchored to the ground or to a wall-mounted bracket with bolt(M10). Prepare the installation base of the unit according to the dimensions below.
- The following is a list of different outdoor unit sizes and the distance between their mounting feet. Prepare the installation base of the unit according to the dimensions below.

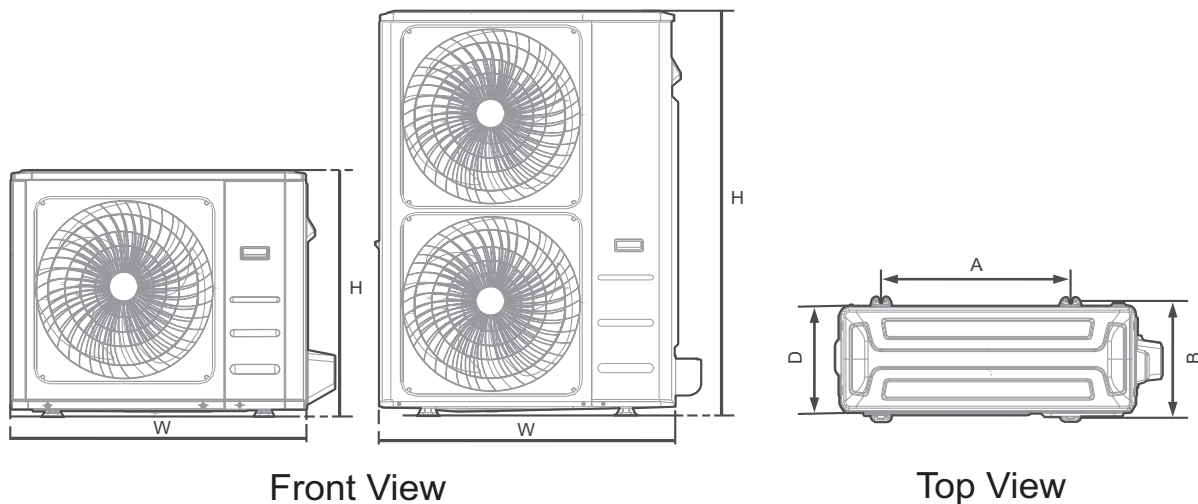


Fig. 13 —Outdoor Unit Types and Specifications (Split Type Outdoor Unit)

Table 6 — Dimensions

Outdoor Size	Mounting Distance	
	Distance A	Distance B
4 and 5 Ton	24.96in (634mm)	15.9in (404mm)
3 Ton	24.25in (616mm)	15.63in (397mm)

CLEARANCES

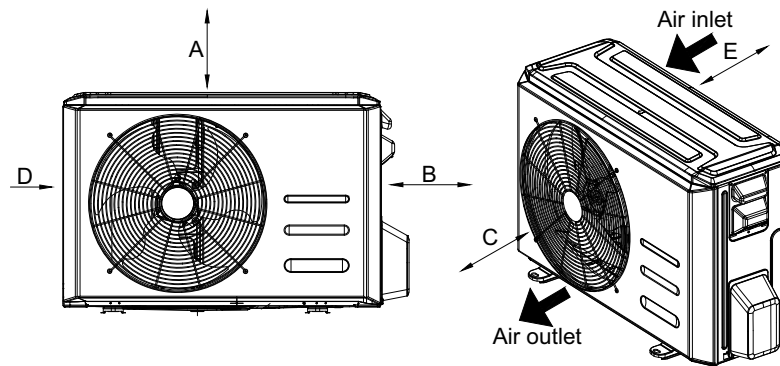


Fig. 14 – Clearances

Table 7 — Clearances

UNIT	MINIMUM VALUE in. (mm)
A	20 (500)
B	14 (350)
C	20 (500)
D	4 (101)
E	4 (101)

NOTE: The outdoor unit must be mounted at least 2in (50mm) above the maximum anticipated snow depth.

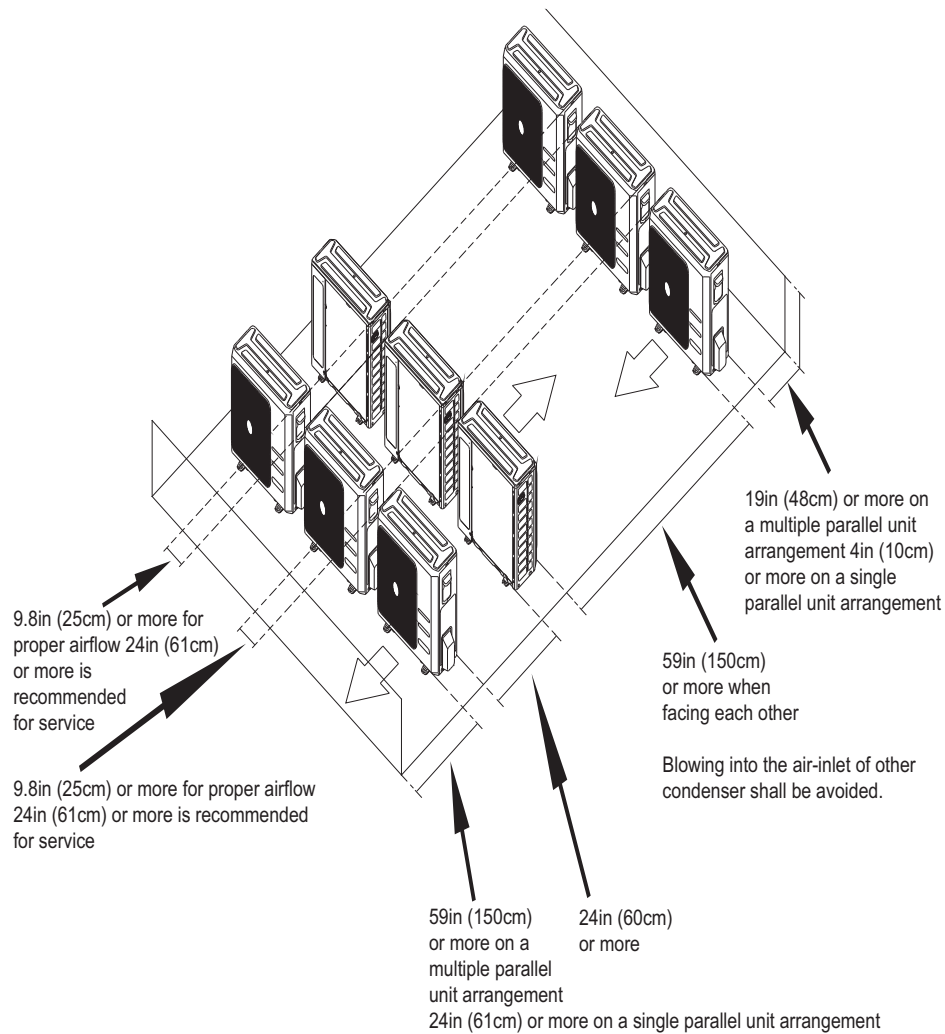


Fig. 15 – Clearances for multiple units

REFRIGERANT CONNECTION INSTRUCTIONS

Use the following steps to connect the refrigerant piping:

1. Run the interconnecting piping from the outdoor unit to the indoor unit.
2. Connect the refrigerant piping and drain line outside the indoor unit. Complete the pipe insulation at the flare connection then fasten the piping and wiring to the wall as required. Completely seal the hole in the wall.
3. Cut tubing to the correct length.

When preparing refrigerant pipes, take extra care to cut and flare them properly. This ensures efficient operation and minimizes the need for future maintenance.

 - a. Measure the distance between the indoor and outdoor units.
 - b. Using a pipe cutter, cut the pipe a little longer than the measured distance.
 - c. Make sure the pipe is cut at a perfect 90° angle.

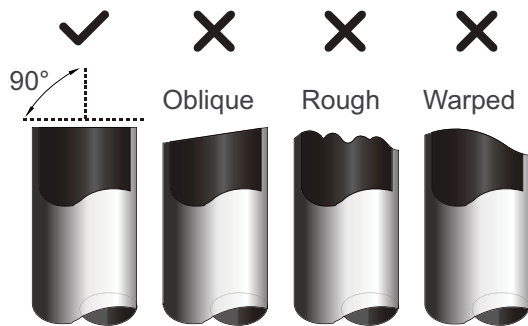


Fig. 16 —Pipe Cutting



CAUTION

DO NOT DEFORM PIPE WHILE CUTTING

Be extra careful not to damage, dent, or deform the pipe while cutting. This will drastically reduce the heating efficiency of the unit.

4. Remove Burrs

Burrs can affect the air-tight seal of the refrigerant piping connection. Therefore, they must be completely removed.

To remove:

 - a. Hold the pipe at a downward angle to prevent burrs from falling into the pipe.
 - b. Using a reamer or deburring tool, remove all burrs from the cut section of the pipe

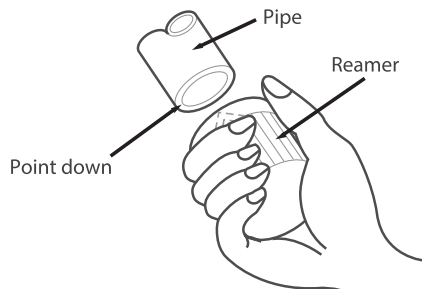


Fig. 17 —Remove Burrs

5. Flare Pipe Ends

Proper flaring is essential to achieve an airtight seal.

- a. After removing burrs from cut pipe, seal the ends with PVC tape to prevent foreign materials from entering the pipe.
- b. Sheath the pipe with insulating material.
- c. Place flare nuts on both ends of pipe. Make sure they are facing in the right direction, because you can not put them on or change their direction after flaring.

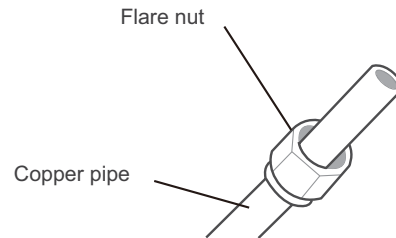


Fig. 18 —Copper Pipe and Flare Nut

- d. Remove PVC tape from ends of pipe when ready to perform flaring work.
- e. Clamp flare form on the end of the pipe. The end of the pipe must extend beyond the flare form.

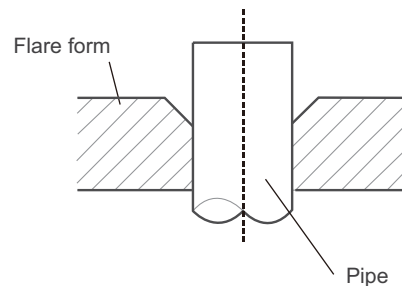


Fig. 19 —Flare Block

- f. Place flaring tool onto the form.
- g. Turn the handle of the flaring tool clockwise until the pipe is fully flared. Flare the pipe in accordance with the dimensions in Table 8.

Table 8 — Flare Nut Spacing

OUTER DIAMETER IN(MM)	"A" IN (MM)	
	MIN.	MAX.
Ø1/4in (Ø6.35mm)	0.0275in (0.7mm)	0.05in (1.3mm)
Ø3/8in (Ø9.52mm)	0.04in (1.0mm)	0.063in (1.6mm)
Ø1/2in (Ø12.7mm)	0.04in (1.0mm)	0.07in (1.8mm)
Ø5/8in (Ø16mm)	0.078in (2.0mm)	0.086in (2.2mm)
Ø3/4in (Ø19mm)	0.078in (2.0mm)	0.094in (2.4mm)

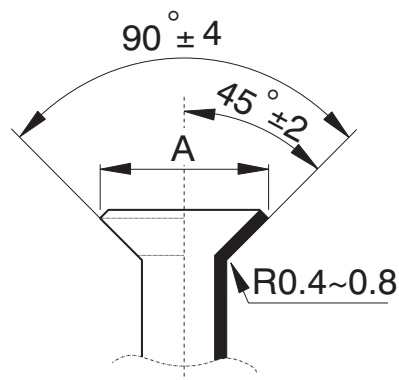


Fig. 20 —Flare Shape

- h. Remove the flaring tool and flare form, then inspect the end of the pipe for cracks and even flaring.
6. Connect Pipes
Connect the copper pipes to the indoor unit first, then connect it to the outdoor unit. Connect the low-pressure pipe first, then the high pressure pipe.
- a. Align the center of the two pipes that you will connect.

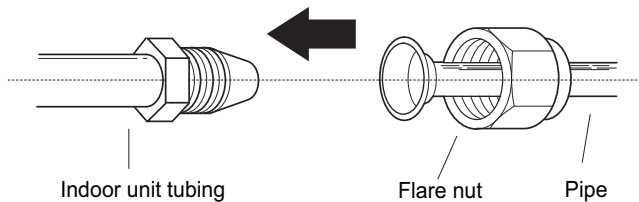


Fig. 21 —Align the Center of the Two Pipes

- b. Tighten the flare nut snugly by hand.
c. Using a wrench, grip the nut on the unit tubing.
d. While firmly gripping the nut, use a torque wrench to tighten the flare nut according to the torque values listed in Table 9.

Table 9 — Tightening Torque

BRASS FLARE SIZES	RECOMMENDED SEATING TORQUE FOR BRASS FLARE NUTS		FLARE DIMENSIONS (A) (INCH/MM)	
	Ft-Lbs	N-M	Min	Max
in (mm)				
Ø1/4 (6.35)	13.3-14.7	18-20	0.33/8.4	0.34/8.7
Ø3/8 (9.52)	23.6-28.8	32-39	0.52/13.2	0.53/13.5
Ø1/2 (12.7)	36.1-42.8	49-59	0.64/16.2	0.65/16.5
Ø5/8 (16)	42-52.4	57-71	0.76/19.2	0.78/19.7
Ø3/4 (19)	49.4-74.5	67-101	0.91/23.2	0.93/23.7
Ø7/8 (22)	62.7-81.1	85-110	1.04/26.4	1.06/26.9

NOTE: Use both a backup wrench and torque wrench when connecting or disconnecting pipes to/from the unit.

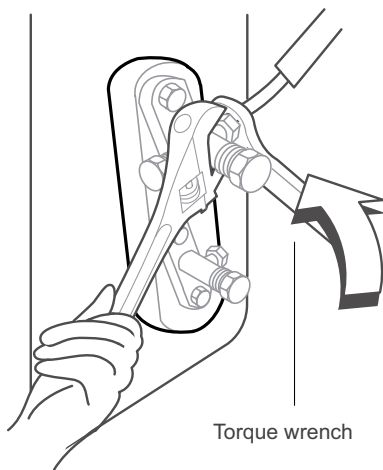


Fig. 22 —Torque Wrench with Backup Wrench

NOTE: MINIMUM BEND RADIUS. Carefully bend the tubing in the middle according to the diagram below. **DO NOT** bend the tubing more than 90° or more than 3 times.

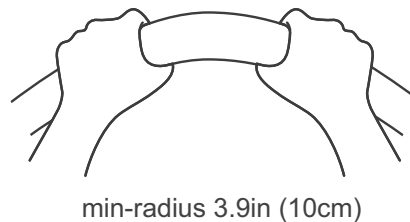


Fig. 23 —Use Care When Bending

- e. After connecting the copper pipes to the indoor unit, wrap the power cable, signal cable and the piping together with binding tape.

NOTE: DO NOT intertwine signal cable with other wires while bundling these items together. **Do not** intertwine or cross the signal cable with any other wiring.

- f. Thread this lineset through the wall and connect it to the outdoor unit.
g. Insulate all piping, including the outdoor unit valves.

NOTE: DO NOT open the service valves until pressure test is complete.

7. Pressure Test Piping



CAUTION

Only use Dry Nitrogen to pressure test refrigerant systems. Use of other gases can result in injury, property damage or death.

NOTE: Use refrigeration gauges that are pressure rated for R454b refrigerant.

- Attach low side gauge hose to the 5/16" Schrader valve on the outdoor unit service valve.
- Attach the charging hose to the regulator on the dry nitrogen tank.
- Preset the nitrogen regulator to 550 psi.
- Slowly pressurize the line set until the low side gauge reads 500 psi. Do not exceed 550 psi.
- Close all the valves on the nitrogen tank and gauges.
- Allow the pressure test to stand for a minimum of 30 minutes.
- If the pressure holds, release the nitrogen and proceed.



CAUTION

Wrap insulation around the piping. Direct contact with the bare piping may result in burns or frostbite. Ensure the pipe is properly connected. Over tightening may damage the bell mouth and under tightening may lead to leakage.

REFRIGERANT PIPING

Table 10 — Piping and Refrigerant Information

OUTDOOR MODEL		37MBRAQ 36AA3	37MBRAQ 48AA3	37MBRAQ 58AA3
Refrigerant Type	Type	R454B		
Charge Amount	lb. (kg)	5.73 (2.6)	7.94 (3.6)	9.03 (4.1)
Additional refrigerant charge	Oz/ft (g/m)	0.32 (30)	0.32 (30)	0.32 (30)
Liquid Pipe (size - connection type)	In (mm)	3/8 (9.52)	3/8 (9.52)	3/8 (9.52)
Suction Pipe (size - connection type)	In (mm)	3/4 (19)	3/4 (19)	3/4 (19)
Min. Piping Length	ft. (m)	10 (3)	10 (3)	10 (3)
Standard Piping Length	ft. (m)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)
Max. Piping Length with no additional refrigerant charge per System	ft. (m)	24.6 (7.5)	24.6 (7.5)	24.6 (7.5)
Total Maximum Piping Length per system	ft. (m)	246 (75)	246 (75)	246 (75)
Max. outdoor-indoor height difference (OU higher than IU)	ft. (m)	98.4 (30)	98.4 (30)	98.4 (30)
Max. outdoor-indoor height difference (IU higher than OU)	ft. (m)	98.4 (30)	98.4 (30)	98.4 (30)

IMPORTANT: Both refrigerant lines must be insulated separately. Use refrigeration grade tubing ONLY. No other type of tubing may be used. Use of other types of tubing will void the manufacturer's warranty.

- All outdoor units have an electronic expansion valve to manage the refrigerant flow of the fan coil connected.
- The minimum refrigerant line length between the indoor and outdoor unit is 10 ft. (3m).
- When paired with Ductless indoor units, size the line sets based on the connection size of the indoor unit unless specified.
- Do not open the service valves or remove the protective caps from the tubing ends until all connections are made.
- Bend the tubing with bending tools to avoid kinks and flat spots.
- Keep the tubing free of dirt, sand, moisture, and other contaminants to avoid damaging the refrigerant system.
- Avoid sags in the suction line to prevent the formation of oil traps.
- Insulate each tube with a minimum 3/8-in. (10 mm) wall thermal pipe insulation. Inserting tubing into the insulation before making the connections will save time and improve installation quality.

**WARNING**

All field piping must be completed by a licensed technician and must comply with the local and national regulations. When the system is installed in a small room, measures must be taken to prevent the refrigerant concentration in the room from exceeding the safety limit in the event of refrigerant leakage. If the refrigerant leaks and its concentration exceeds its proper limit, hazards due to lack of oxygen may result. When installing the refrigeration system, ensure that air, dust, moisture or foreign substances do not enter the refrigerant circuit. Contamination in the system may cause poor operating capacity, high pressure in the refrigeration cycle, explosion or injury. Ventilate the area immediately if there is refrigerant leakage during the installation. Leaked refrigerant gas is hazardous. Ensure there is no refrigerant leakage after completing the installation work.

**CAUTION**

DO NOT install the connecting pipe until both the indoor and outdoor units have been installed. Insulate both the gas and liquid piping to prevent condensation.

WIRING INSTRUCTIONS**WIRING PRECAUTIONS****WARNING**

Before performing any electrical work, read the following bullet points.

- All wiring must comply with local and national electrical codes, regulations and must be installed by a licensed electrician.
- All electrical connections must be made according to the Electrical Connection Diagram located on the panels of the indoor and outdoor units.
- If there is a serious safety issue with the power supply, stop work immediately. Explain your reasoning to the client, and refuse to install the unit until the safety issue is properly resolved.
- Power voltage should be within 90-110% of rated voltage. Insufficient power supply can cause malfunction, electrical shock, or fire.
- Installation of an external surge suppressor at the outdoor disconnect is recommended.
- If connecting power to fixed wiring, a switch or circuit breaker that disconnects all poles and has a contact separation of at least 1/8in (3mm) must be incorporated in the fixed wiring. The qualified technician must use an approved circuit breaker or switch.
- Only connect the unit to an individual branch circuit. Do not connect another appliance to that outlet.
- Make sure to properly ground the unit.
- Every wire must be firmly connected. Loose wiring can cause the terminal to overheat, resulting in product malfunction and possible fire.
- Do not let wires touch or rest against refrigerant tubing, the compressor, or any moving parts within the unit.

- To avoid getting an electric shock, never touch the electrical components soon after the power supply has been turned off. After turning off the power, always wait 10 minutes or more before you touch the electrical components.
- Make sure that you do not cross your electrical wiring with your signal wiring. This may cause distortion, interference or possibly damage to circuit boards.
- No other equipment should be connected to the same power circuit.
- Connect the outdoor wires before connecting the indoor wires.



WARNING

Before performing any electrical or wiring work, turn off the main power to the system.

CONNECTION DIAGRAMS

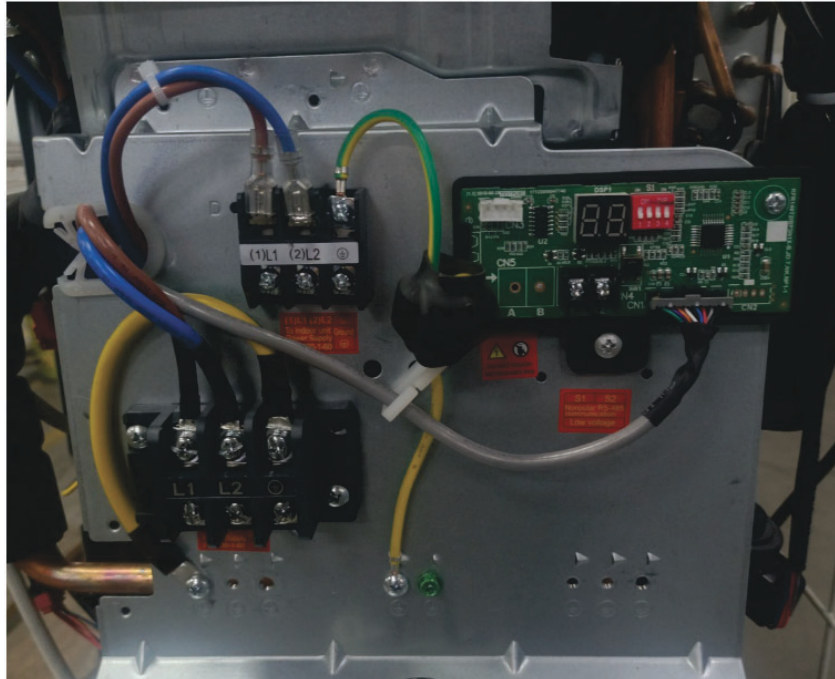


Fig. 24 —Grounding Wire Image - Size 36K

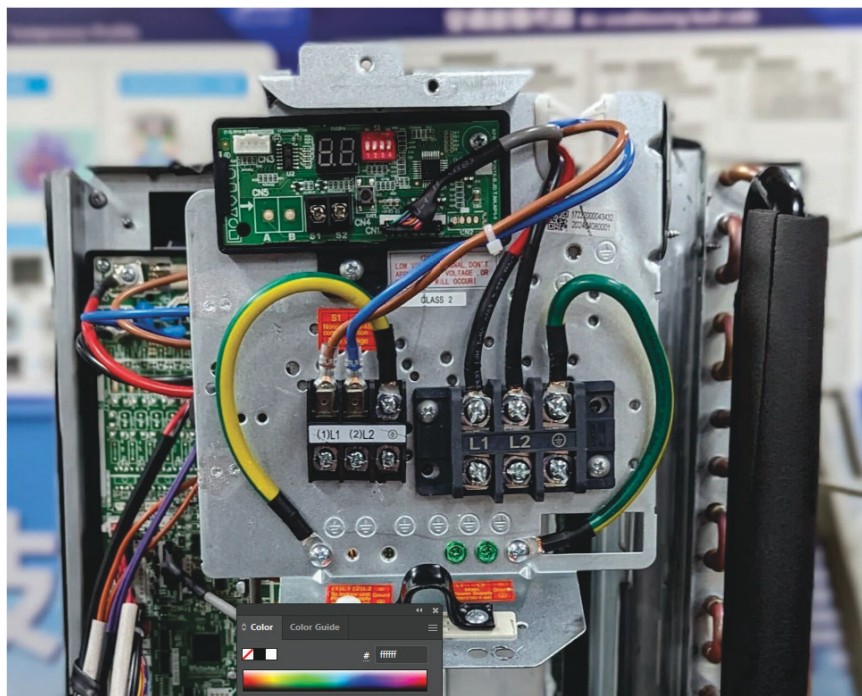


Fig. 25 —Grounding Wire Image - Size 48K-58K

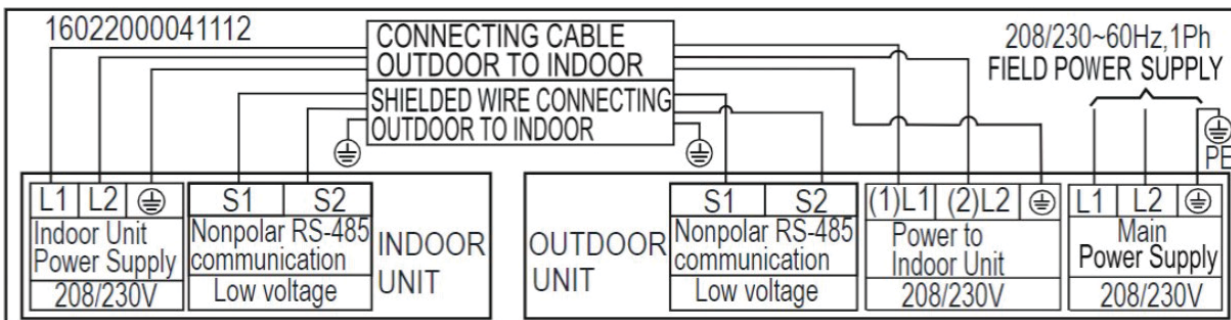


Fig. 26 —Connection Diagram (Sizes 36K-58K)

NOTES:

1. Do not use the thermostat wire for any connection between indoor and outdoor units.
2. All connections between indoor and outdoor units must be as shown in Figure 26. The connections are sensitive to polarity and will result in a fault code.

NOTE ON CIRCUIT BREAKER: When the maximum current of the air conditioner is more than 16A, a circuit breaker or leakage protection switch with protective device shall be used (purchased separately). The unit should be wired according to NEC and CEC requirements.

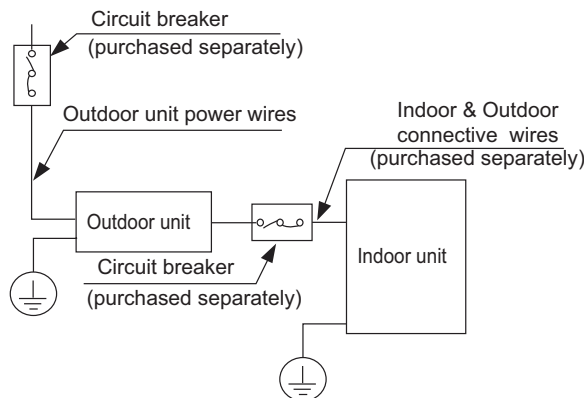


Fig. 27 —Circuit Breaker Wiring

NOTE: The images are for explanation purpose only. Your unit may be slightly different.

OUTDOOR UNIT WIRING**WARNING**

Before performing any electrical or wiring work, turn off the main power to the system.

1. Prepare the cable for connection.
 - a. You must first choose the right cable size. Choose the cable type according to the local electrical codes and regulations.
 - b. The size of the power supply cable, signal cable, fuse, and switch needed is determined by the Minimum Circuit Ampacity of the unit. The Minimum Circuit Ampacity is indicated on the nameplate located on the side panel of the unit. Refer to this nameplate to choose the right cable, fuse, or switch.
 - c. Using wire strippers, strip the rubber jacket from both ends of the signal cable to reveal approximately 5.9in (15cm) of wire.
 - d. Strip the insulation from the ends.
 - e. Using a wire crimper, crimp u-lugs on the ends.

NOTE: When connecting the wires, strictly follow the wiring diagram found inside the electrical box cover.

2. Remove the 2 screws on the front and side panels. Take it down to perform wire connection. See Fig. 28.
Unscrew the electrical wiring cover and remove it. See Fig 28.

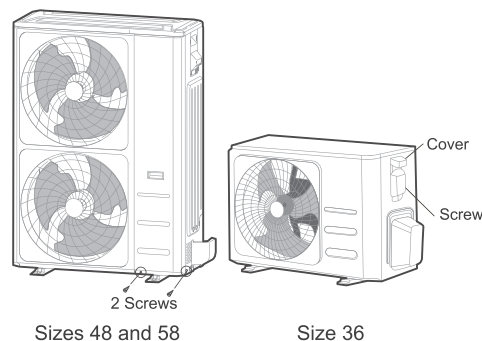


Fig. 28 —Remove Panels and Electrical Wiring Cover

3. Connect the u-lugs to the terminals. Match the wire colors/labels with the labels on the terminal block. Firmly screw the u-lug of each wire to its corresponding terminal.
4. Clamp down the cable with the cable clamp.

5. Insulate unused wires with electrical tape. Keep them away from any electrical or metal parts.
6. Reinstall the cover of the electric control box.

ELECTRICAL CONNECTIONS

Install All Power and Interconnecting Wiring to Outdoor Units

1. Mount the outdoor power disconnect.
2. Run the power wiring from the main box to disconnect per NEC and local codes.
3. Remove the field wiring cover from the unit by loosening the screws.
4. Remove the caps on the conduit panel.
5. Connect the conduit to conduit panel (see Fig.29).
6. Properly connect both the power supply and control lines to the terminal block per the connection diagram for the appropriate unit capacity and voltage.
7. Ground the unit in accordance with NEC and local electrical codes.
8. Use the lock nuts to secure the conduit.
9. Reinstall the field wiring cover.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

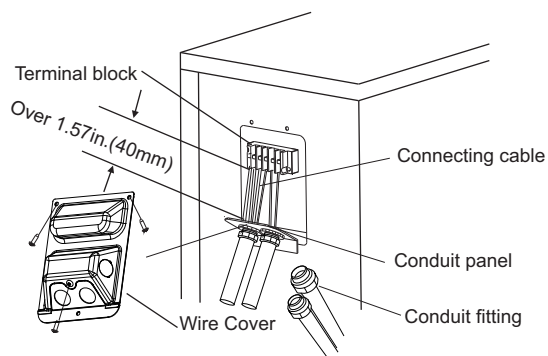
Be sure to comply with local codes while running wire from indoor unit to outdoor unit.

Every wire must be connected firmly. Loose wiring may cause the terminal to overheat or result in unit malfunction. A fire hazard may also exist. Therefore, ensure all wiring is tightly connected.

No wire should be allowed to touch the refrigerant tubing, compressor or any moving parts.

Disconnecting means must be provided and shall be located within sight and readily accessible from the air conditioner.

Connecting cable with the conduit shall be routed through hole in the conduit panel.



Select the appropriate conduit opening for the wire size used.

Fig. 29 —Field Wiring



WARNING

An approved and listed fitting must be used to securely affix conduit in accordance with NEC and local codes.

WIRING INSTRUCTIONS

The main power is supplied to the outdoor unit as shown in Table 12. All wires must be sized per NEC (National Electrical Code) or CEC (Canadian Electrical Code) and local codes. Use Electrical Data table MCA (minimum circuit amps) and MOCP (maximum over current protection) to correctly size the wires and the disconnect fuse or breakers respectively.

SIZES 36-58 RECOMMENDED CONNECTION METHOD FOR POWER AND COMMUNICATION WIRING

Power and Communication Wiring: The field supplied power wiring from the outdoor unit to the indoor unit consists of three (3) wires and provides the power for the indoor unit. Two wires are high voltage AC power and one is a ground wire. To minimize voltage drop, the factory recommended wire size is 14/2 stranded with a ground.

Communication Wiring: A separate shielded stranded copper conductor only, with a 600 volt rating and double insulated copper wire, must be used as the communication wire from the outdoor unit to the indoor unit.

Please use a separate shielded 16GA stranded control wire. See Fig. 30.

Table 11 — Wiring Connections

System Size	36K	48K	58K
V-Ph-Hz	208/230-1-60		
Power Supply	3Wires: includes ground wire (Outdoor)		
Interconnection to Indoor Unit	Yes		
Shielded Wire	Yes		



WARNING

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. Wires should be sized based on NEC and local codes.



CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. Be sure to comply with local codes while running wire from the indoor unit to the outdoor unit. Every wire must be connected firmly. Loose wiring may cause the terminal to overheat or result in unit malfunction. A fire hazard may also exist. Ensure all wiring is tightly connected.

No wire should touch the refrigerant tubing, compressor or any moving parts. Disconnecting means must be provided and shall be located within sight and readily accessible from the air conditioner. Connecting cable with conduit shall be routed through the hole in the conduit panel.

ELECTRICAL DATA

Table 12 — Electrical Data

Outdoor Unit Size		36K	48K	58K
Power (Voltage-Phase-Frequency)	V-P-H	208/230-1-60		
MCA (Minimum Circuit Ampacity)	A	33	39	42
MOPA (Maximum Overcurrent Protection Ampacity)	A	35	40	45
Min – Max Voltage Range	V	187-253		
Cooling (with Ceiling and Floor IDU)				
Running current	A	15.3	22	24.6
Power consumption	W	3529	5053	5625
Heating (with Ceiling and Floor IDU)				
Running current range	A	14.3	22.6	26.2
Power consumption	W	3334	5196	5765

LEGEND
MCA - Minimum Circuit Amps
MOCP - Maximum Over-Current Protection

WIRING DIAGRAMS

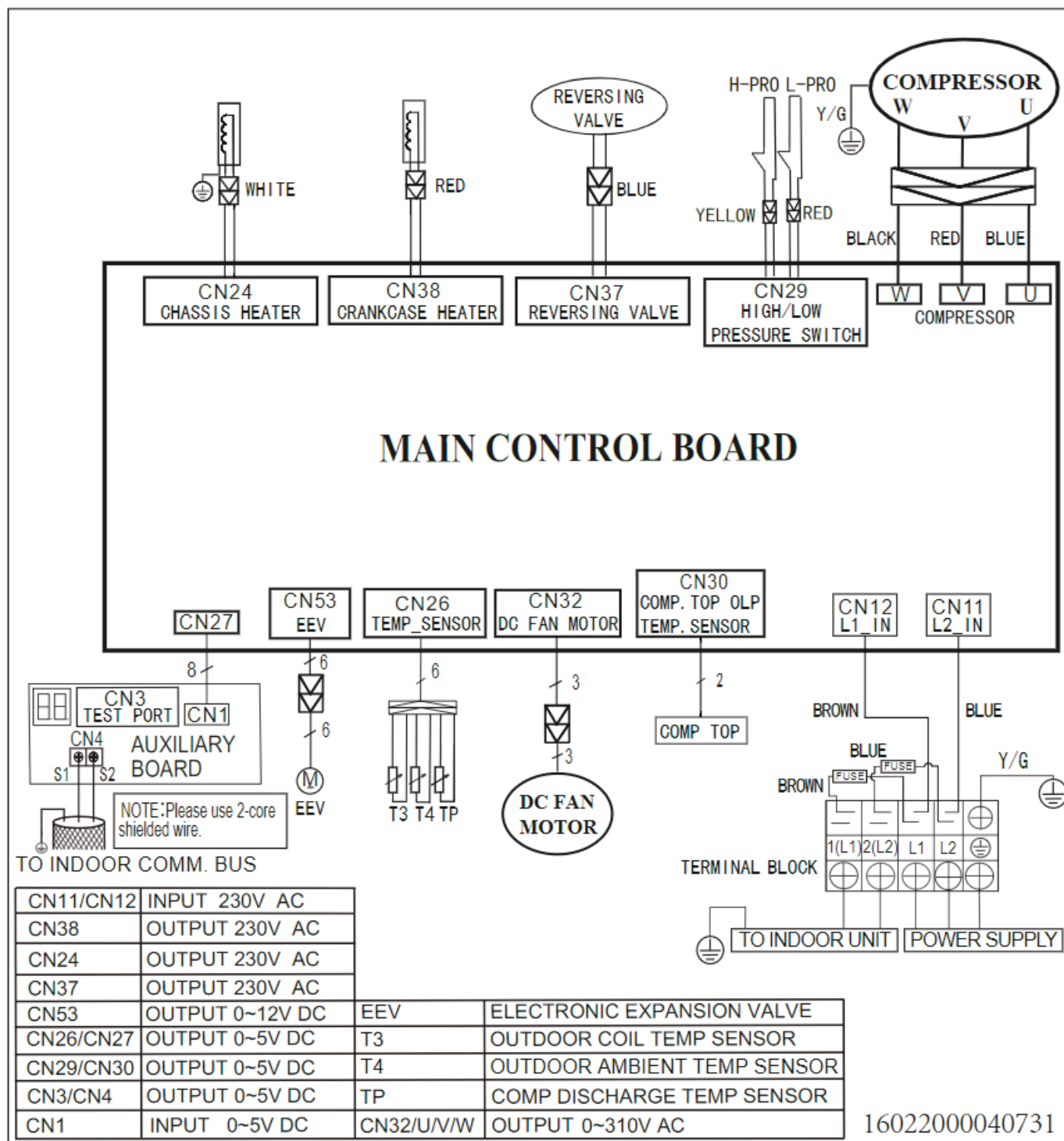


Fig. 30 —Wiring Diagram - Size 36K

NOTE: Refer to the actual diagram on the unit.

WIRING DIAGRAMS (CONT.)

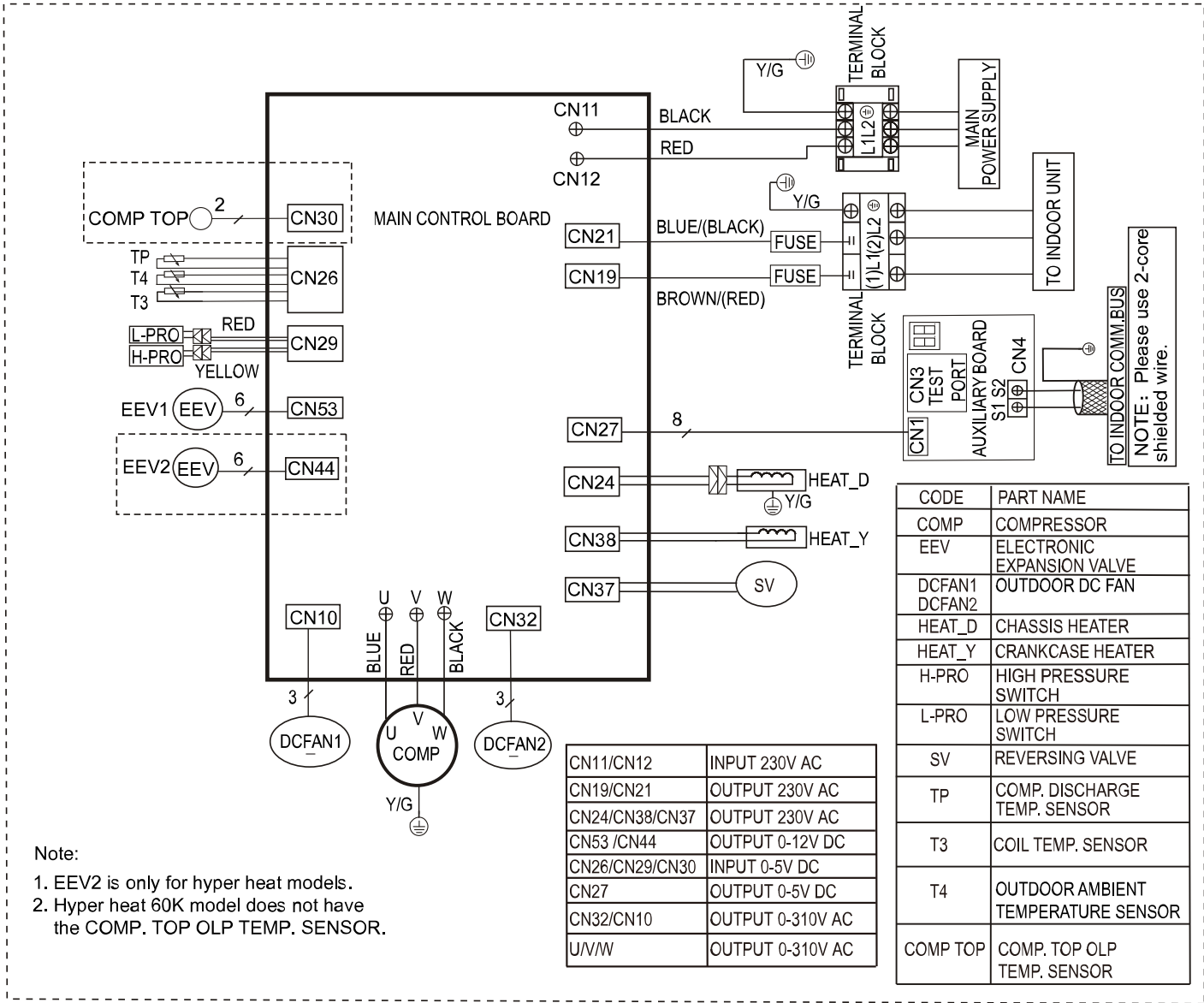


Fig. 31 —Wiring Diagram - Size 48K-58K

NOTE: Refer to the actual diagram on the unit.

AIR EVACUATION

CAUTION

The service port (Schrader Fitting) for R454b refrigerant is a special fitting that requires a new type adapter.

The Service Port Adapter used for R410A refrigerant units will not work with the new equipment.

This new adapter uses Reverse threads (Left-Hand Thread) and must be obtained from distributor or supply house.

CAUTION

UNIT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Never use the system compressor as a vacuum pump.

NOTE: When opening valve stems, turn the hexagonal wrench until it hits against the stopper. Do not try to force the valve to open further.

PREPARATION AND PRECAUTION

Air and foreign matter in the refrigerant circuit can cause abnormal rises in pressure, which can damage the air conditioner, reduce its efficiency,

and cause injury. Use a vacuum pump and manifold gauge to evacuate the refrigerant circuit, removing any non-condensable gas and moisture from the system. Evacuation should be performed upon initial installation and when unit is relocated.

BEFORE PERFORMING EVACUATION

- ☑ Check to make sure the connective pipes between the indoor and outdoor units are connected properly.
- ☑ Check to make sure all wiring is connected properly.

EVACUATION INSTRUCTIONS

1. Connect the charge hose of the manifold gauge to service port on the outdoor unit's low pressure valve.
2. Connect another charge hose from the manifold gauge to the vacuum pump.
3. Open the Low Pressure side of the manifold gauge. Keep the High Pressure side closed.
4. Turn on the vacuum pump to evacuate the system.
5. Run the vacuum for at least 15 minutes, or until the Compound Meter reads 28 inches of mercury or 500 microns.

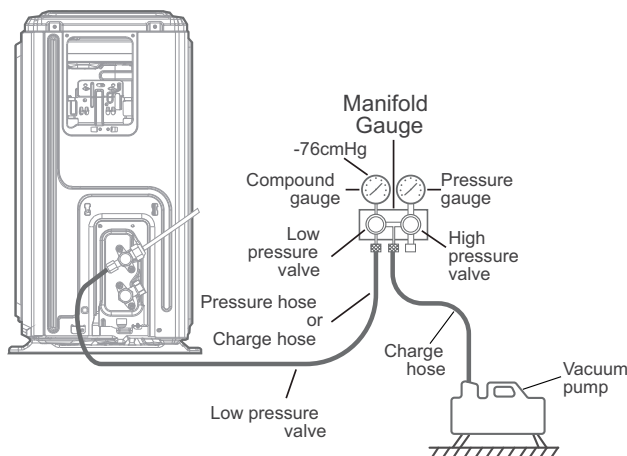


Fig. 32 —Evacuation Instructions - Outdoor Unit

6. Close the Low Pressure side of the manifold gauge, and turn off the vacuum pump.
7. Wait for 5 minutes, then check that there has been no change in system pressure.
8. If there is a change in system pressure, refer to Gas Leak Check section for information on how to check for leaks. If there is no change in system pressure, unscrew the cap from the packed valve (high pressure valve).
9. Insert hexagonal wrench into the packed valve (high pressure valve) and open the valve by turning the wrench in a 1/4 counterclockwise turn. Listen for gas to exit the system, then close the valve after 5 seconds.
10. Watch the Pressure Gauge for one minute to make sure that there is no change in pressure. The Pressure Gauge should read slightly higher than atmospheric pressure.
11. Remove the charge hose from the service port.

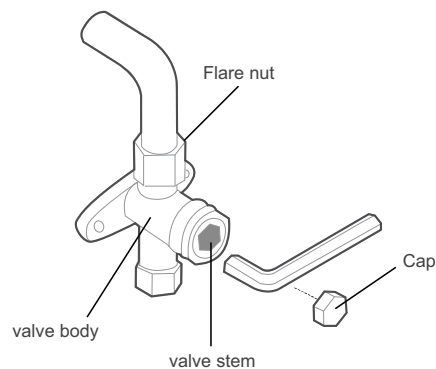


Fig. 33 —Evacuation Instructions

12. Using hexagonal wrench, fully open both the high pressure and low pressure valves.
13. Tighten valve caps on all three valves (service port, high pressure, low pressure) by hand. You may tighten it further using a torque wrench if needed.

ADDING REFRIGERANT



CAUTION

NO NOT MIX REFRIGERANT TYPES.

Some systems require additional charging depending on pipe lengths. In North America, the standard pipe length is 25ft (7.5m). The refrigerant should be charged from the service port on the outdoor unit's low pressure valve. The additional refrigerant to be charged can be calculated using the following formula (the refrigerant quantity added in the following table is a reference value and can be adjusted according to the actual situation):

Table 13 — Adding R454B Refrigerant

LIQUID SIDE DIAMETER	Ø3/8in (Ø9.52mm)
PIPE LENGTH	(Standard length) x 0.32oz/ft (Standard length) x 30g/m

DEEP VACUUM METHOD

The deep vacuum method requires a vacuum pump capable of pulling a vacuum of 500 microns and a vacuum gauge capable of accurately measuring this vacuum depth. The deep vacuum method is the most positive way of assuring a system is free of air and moisture (see Figure 34).

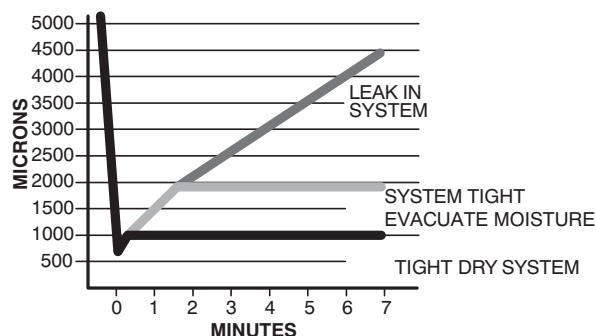


Fig. 34 — Deep Vacuum Graph

TRIPLE EVACUATION METHOD

The triple evacuation method should be used when vacuum pump is not capable of pumping down to 500 microns and system does not contain any liquid water. Refer to Fig. 35 and proceed as follows:

1. Attach refrigeration gauges and evacuate system down to 28 in. of mercury and allow pump to continue operating for an additional 15 minutes.
2. Close service valves and shut off vacuum pump.
3. Connect a nitrogen cylinder and regulator to system and flow nitrogen until system pressure is 2 psig.
4. Close service valve and allow system to stand for 1 hour. During this time, dry nitrogen will be able to diffuse throughout the system absorbing moisture.
5. Repeat this procedure as indicated in Fig. 35. System will then be free of any contaminants and water vapor.

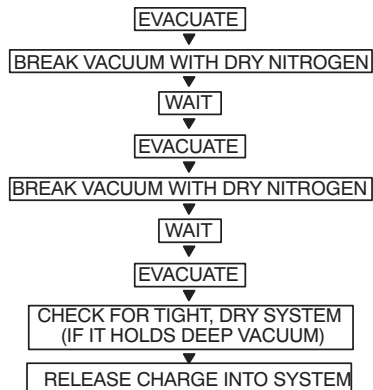


Fig. 35 — Triple Evacuation Method

TEST RUN

CAUTION

Failure to perform the test run may result in unit damage, property damage, or personal injury.

BEFORE TEST RUN

A test run must be performed after the entire system has been completely installed. Confirm the following points before performing the test:

1. Indoor and outdoor units are properly installed.
2. Piping and wiring are properly connected.
3. No obstacles near the inlet and outlet of the unit that might cause poor performance or product malfunction.
4. Refrigeration system does not leak.
5. Drainage system is unimpeded and draining to a safe location.
6. Heating insulation is properly installed.
7. Grounding wires are properly connected.
8. Length of the piping and additional refrigerant capacity have been recorded.
9. Power voltage is the correct voltage for the air conditioner

TEST RUN INSTRUCTIONS

1. Open both the liquid and gas stop valves.
2. Turn on the main power switch and allow the unit to warm up.
3. Set the air conditioner to COOL mode.
4. For the Indoor Unit
 - a. Ensure the remote control and its buttons work properly.
 - b. Ensure the louvers move properly and can be changed using the remote control.
 - c. Double check to see if the room temperature is being registered correctly.
 - d. Ensure the indicators on the remote control and the display panel on the indoor unit work properly.

- e. Ensure the manual buttons on the indoor unit works properly.
 - f. Check to see that the drainage system is unimpeded and draining smoothly.
 - g. Ensure there is no vibration or abnormal noise during operation.
5. For the Outdoor Unit
 - a. Check to see if the refrigeration system is leaking.
 - b. Make sure there is no vibration or abnormal noise during operation.
 - c. Ensure the wind, noise, and water generated by the unit do not disturb your neighbors or pose a safety hazard.
 6. Drainage Test
 - a. Ensure the drainpipe flows smoothly. New buildings should perform this test before finishing the ceiling.
 - b. Remove the test cover. Add 2,000ml of water to the tank through the attached tube.
 - c. Turn on the main power switch and run the air conditioner in COOL mode.
 - d. Listen to the sound of the drain pump to see if it makes any unusual noises.
 - e. Check to see that the water is discharged. It may take up to one minute before the unit begins to drain depending on the drainpipe.
 - f. Make sure that there are no leaks in any of the piping.
 - g. Stop the air conditioner. Turn off the main power switch and reinstall the test cover.

NOTE: If the unit malfunctions or does not operate according to your expectations, please refer to the Troubleshooting section of the Owner's Manual before calling customer service.

CARE AND MAINTENANCE

To help ensure high performance and minimize possible equipment failure, periodic maintenance must be performed on this equipment. Maintenance frequency may vary depending upon geographic areas.

TROUBLESHOOTING

For ease of service, the systems are equipped with diagnostic code display LEDs on both the indoor and outdoor units. The outdoor diagnostic display has one red LED on the outdoor unit board and is limited to very few errors. The indoor diagnostic display is a combination of flashing LEDs on the display panel or the front of the unit. There may be a few error codes displayed in the indoor unit that might relate to the outdoor unit's problems. If possible, always check the diagnostic codes displayed on the indoor unit first.

Table 14 — Error Codes

Display	Malfunction and Protection Indication	Display	Malfunction and Protection Indication
EC01	Other indoor unit refrigerant sensor detects a leak (multi-zone)	PC08	ODL Current Protection
EC07	Outdoor DC fan motor speed out of control	PC10	ODL AC voltage too low protection
EC51	ODU EEPROM error	PC11	ODL DC bus voltage too high protection
EC52	Condenser coil temperature sensor (T3) error	PC12	ODL DC bus voltage too low protection
EC53	ODU temperature sensor (T4) error	PC30	System pressure overload protection
EC54	ODU exhaust temperature sensor error	PC31	System pressure too low protection
EC55	ODU IPM module temperature sensor error	PC40	Communication failure between outdoor main control chip and the driver chip
EC56	ODU T2B sensor	PC41	Compressor current sampling circuit failure
EH00	IDU EEPROM error	PC42	Compressor starting failure
EH0A	Indoor EEPROM Parameter error	PC43	Compressor lost phrase protection
EH01	Refrigerant Sensor detects a leak	PC44	Compressor zero speed protection
EH02	Working condition of the refrigerant sensor is out of range and a leak is detected	PC45	Voltage drop
EH03	Working condition of the refrigerant sensor is out of range	PC46	Compressor speed out of control
EH02	Zero-crossing signal detection error	PC49	Compressor over current error
EH03	Indoor fan motor speed is out of control	PC0A	Condenser high temperature protection
EH31	Protection for low DC bus voltage of the external fan	PC0F	PFC failure
EH32	Protection for high DC bus voltage of the external fan	PC0L	Outdoor low temperature protection
EH60	IDU ENV temperature T1 sensor error	PH09	IDU anti-cold wind stop machine
EH61	IDU pipe temperature T2 sensor error	PH90	Evaporator high temperature protection
EH0b	IDU PCB and display communication error	PH91	Evaporator low temperature protection
FH0C	Indoor Unit humidity sensor malfunction	LC01	Condenser high temperature frequency limited (L1)
FHCC	Refrigerant Sensor Error	LC02	Compressor Discharge Pipe High temperature frequency limited (L2)
EL01	IDU and ODU Communication Error	LC03	Current frequency limited (L3)
EL0C	System leaks refrigerant	LC05	Voltage frequency limited (L5)
FH0P	Wireless Module Self-Test Failure	LC06	IPM module temperature frequency limited
FL09	New and old platform mismatch failure	LH00	Evaporator temperature frequency limited (L0)
PC00	ODU IPM Protection	LH07	Remote Control frequency limitation in effect
PC01	ODU Voltage Protection		
PC02	Compressor top temperature (IPM module temperature protection)	---	Mode conflict fault
		nA	No fault or protection
PC03	System Pressure Protection		

Table 15 — Refrigerant Leak Detection Error Codes

EH01	Refrigerant Sensor detects a leak
EH02	Working condition of the refrigerant sensor is out of range and a leak is detected

If you receive one of the codes listed in Table 15, call a technician as soon as possible. No need to panic, the unit goes into **TURBO** mode until the error code is cleared. There is a “beep” noise coming from the indoor unit, which is normal in this case. For additional diagnostic information, refer to the service manual.

DUCTLESS START-UP CHECKLIST

Installation Data

Site Address: _____

City: _____ State: _____ Zip Code: _____

Installing Contractor: _____ Contractor Contact #: () _____ - _____

Job Name: _____ Start-up Date: _____

Distributor: _____

System Details

Units	Model No.	Serial No.	Controller
OUTDOOR UNIT			
INDOOR UNIT A			

Wiring Electrical

Wire Size and Type Used? AWG: _____ TYPE: _____

Are there any breaks, splices, wire nuts or butt connectors between the outdoor unit and the indoor door unit? YES: _____ NO: _____

Was the wiring from the outdoor unit port to the correct indoor unit verified? YES: _____ NO: _____

REMARKS: _____

Voltage Check

Wiring: Single zone _____

Outdoor Unit Disconnect	1(L1):GND		Outdoor Unit Terminal Block	1(L1):GND		NOTES: _____ _____ _____ _____
	2(L2):GND			2(L2):GND		
	1(L1):L2(2)			1(L1):2(L2)		
Indoor Unit Voltage Check @ Outdoor Unit	1(L1):GND		Indoor Unit Voltage Check @ Indoor Unit	1(L1):GND		NOTES: _____ _____ _____ _____
	2(L2):GND			2(L2):GND		
	1(L1):2(L2)			1(L1):2(L2)		
	2(L2):3(S)			2(L2):3(S)		

Piping

Leak Check:

System held 500 psig (max. 550psi) for a minimum of 30 minutes using dry nitrogen. YES: _____ NO: _____

Evacuation Method:

- Was the Triple Evacuation Method used as outlined in the installation manual? YES: _____ NO: _____
- Was the Deep Vacuum Method used as outlined in the installation manual? YES: _____ NO: _____
- Did the System Hold 500 microns for 1 hour? YES: _____ NO: _____
- Does the line set match the diameter of the evaporator connections? YES: _____ NO: _____

Ductless Start-Up Checklist (CONT)**Single Zone Piping:**

Has the liquid pipe length been measured and the additional charge calculated? Size: _____ Length: _____ Charge: _____

Notes:

PORT	LIQUID SIZE		SUCTION SIZE		LENGTH	CHARGE	NOTES:
A							

Performance Check

For 1:1 Single Zone Systems: Adjust the set-point to create an operational call for the desired testing operation. Allow the system to run for a minimum of 10 min. and record the following details:

(Operational data recorded on applicable heads with the wireless remote controller's Point Check function)

UNIT	SET-POINT	MODE	T1	T2	T3	T4	Tb	Tp	Th	LA/Lr
A										

Error Codes

Were there any error codes present at start-up?

YES: _____ NO: _____

Indoor Unit Error Code:		Notes:
Outdoor Unit Error Code:		
Wall Controller:		
24V Interface:		

Comments:
